

Unit 1 AP Topics 15

1. A particle moving along a curve in the xy -plane has position $(x(t), y(t))$, at time $t > 0$, where

$$\frac{dx}{dt} = \left(\frac{6}{t} - 3\right)^{1/3} \text{ and } \frac{dy}{dt} = te^{-t}.$$

At time $t = 3$, the particle is at the point $(5, 4)$.

Describe the behavior of the path of the particle as t increases without bound.

Part D

One point is earned for the behavior of x

One point is earned for the behavior of y

Since $\lim_{x \rightarrow \infty} \frac{dx}{dt} = (-3)^{1/3}$ and $\lim_{x \rightarrow \infty} \frac{dy}{dt} = 0$,

the particle will continue moving farther to the left while approaching a horizontal asymptote.



0	1	2
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The student earns all of the following points:

One point is earned for the behavior of x

One point is earned for the behavior of y

Since $\lim_{x \rightarrow \infty} \frac{dx}{dt} = (-3)^{1/3}$ and $\lim_{x \rightarrow \infty} \frac{dy}{dt} = 0$,

the particle will continue moving farther to the left while approaching a horizontal asymptote.

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2.  A particle moving along a curve in the xy -plane has position $(x(t), y(t))$, at time $t > 0$, where

$$\frac{dx}{dt} = \left(\frac{6}{t} - 3\right)^{1/3} \text{ and } \frac{dy}{dt} = te^{-t}.$$

At time $t = 3$, the particle is at the point $(5, 4)$.

- Find the speed of the particle at time $t = 3$.
- Write an equation for the line tangent to the path of the particle at time $t = 3$.
- Is there a time t at which the particle is farthest to the right? If yes, explain why and give the value of t and the x -coordinate of the position of the particle at that time. If no, explain why not.
- Describe the behavior of the path of the particle as t increases without bound.

Part A

1 point is earned for the answer

$$\text{Speed} = \sqrt{(x'(3))^2 + (y'(3))^2} = 1.011$$



0	1
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The student earns all of the following points:

1 point is earned for the answer

$$\text{Speed} = \sqrt{(x'(3))^2 + (y'(3))^2} = 1.011$$

Part B

1 point is earned for the slope

1 point is earned for the equation of tangent line

$$\text{At } t = 3, \frac{dy}{dx} = \frac{y'(3)}{x'(3)} = \frac{3e^{-3}}{-1} = -\frac{3}{e^3} = -0.149$$

An equation for the line tangent to the path at the point $(5, 4)$ is $y - 4 = -\frac{3}{e^3}(x - 5)$.

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0	1	2
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The student earns all of the following points:

1 point is earned for the slope

1 point is earned for the equation of tangent line

$$\text{At } t = 3, \frac{dy}{dx} = \frac{y'(3)}{x'(3)} = \frac{3e^{-3}}{-1} = -\frac{3}{e^3} = -0.149$$

An equation for the line tangent to the path at the point (5, 4) is $y - 4 = -\frac{3}{e^3}(x - 5)$.

Part C

1 point is earned for "yes" with explanation

1 point is earned for $t = 2$

1 point is earned for integral for $x(2)$

1 point is earned for x -coordinate

Since $\frac{dx}{dt} > 0$ for $0 < t < 2$ and $\frac{dx}{dt} < 0$ for $t > 2$, the particle is farthest to the right at $t = 2$. The x -coordinate

$$\text{of the position is } x(2) = 5 + \int_3^2 \left(\frac{6}{t} - 3 \right)^{1/3} dt = 5.791 \text{ or } 5.792$$



0	1	2	3	4
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The student earns all of the following points:

1 point is earned for "yes" with explanation

1 point is earned for $t = 2$

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1 point is earned for integral for $x(2)$

1 point is earned for x -coordinate

Since $\frac{dx}{dt} > 0$ for $0 < t < 2$ and $\frac{dx}{dt} < 0$ for $t > 2$, the particle is farthest to the right at $t = 2$. The x -coordinate

of the position is $x(2) = 5 + \int_3^2 \left(\frac{6}{t} - 3 \right)^{1/3} dt = 5.791$ or 5.792

Part D

1 point is earned for the behavior of x

1 point is earned for the behavior of y

Since $\lim_{t \rightarrow \infty} \frac{dx}{dt} = (-3)^{1/3}$ and $\lim_{t \rightarrow \infty} \frac{dy}{dt} = 0$, the particle will continue moving farther to the left while approaching a horizontal asymptote.



0	1	2
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The student earns all of the following points:

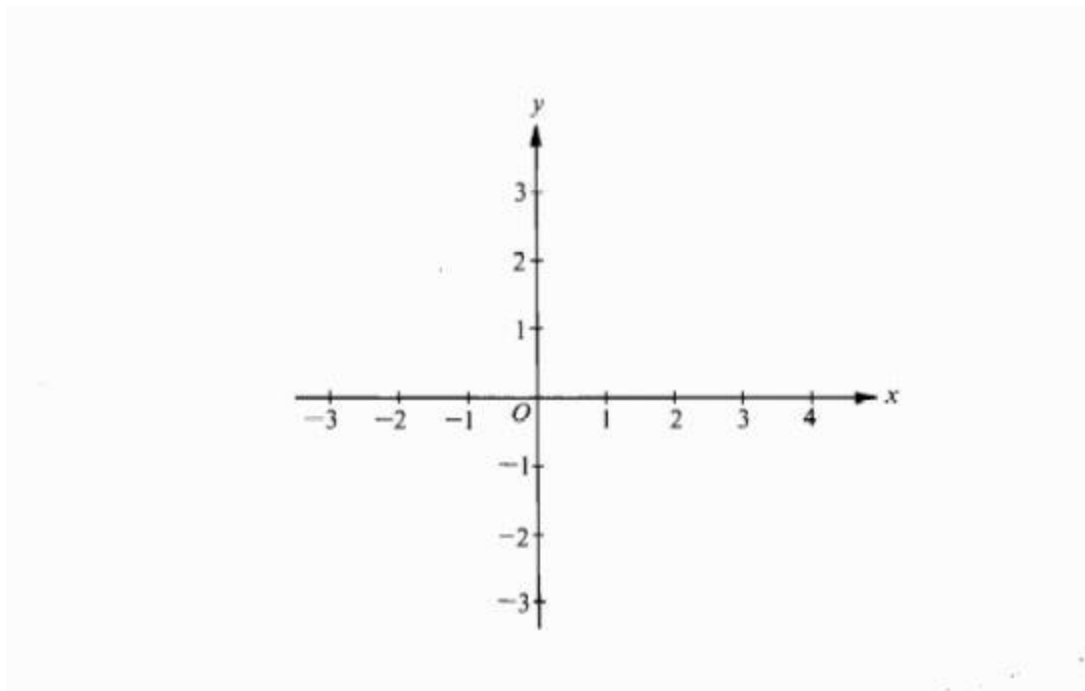
1 point is earned for the behavior of x

1 point is earned for the behavior of y

Since $\lim_{t \rightarrow \infty} \frac{dx}{dt} = (-3)^{1/3}$ and $\lim_{t \rightarrow \infty} \frac{dy}{dt} = 0$, the particle will continue moving farther to the left while approaching a horizontal asymptote.

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3. Let f be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .
- (a) Write an equation of the horizontal asymptote for the graph of f .
- (b) Find the x -coordinate of each critical point of f . For each such x , determine whether $f(x)$ is a relative maximum, a relative minimum, or neither.
- (c) For what values of x is the graph of f concave down?
- (d) Using the results found in parts (a), (b), and (c), sketch the graph of $y = f(x)$ in the xy -plane provided below.

**Part A**

1 point is earned for the correct equation

$$y = 0$$



0

1

The student response earns all of the following points:

1 point is earned for the correct equation

$$y = 0$$

Part B

1 point is earned for correct derivative

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1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x = 1$

relative maximum at $x = 1$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x = 1$

relative maximum at $x = 1$

Part C

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1 - x) \\ &= 2e^{-x}(x - 2) \end{aligned}$$

Concave down when

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$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$



0	1	2
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The student response earns all of the following points:

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\ &= 2e^{-x}(x-2) \end{aligned}$$

Concave down when

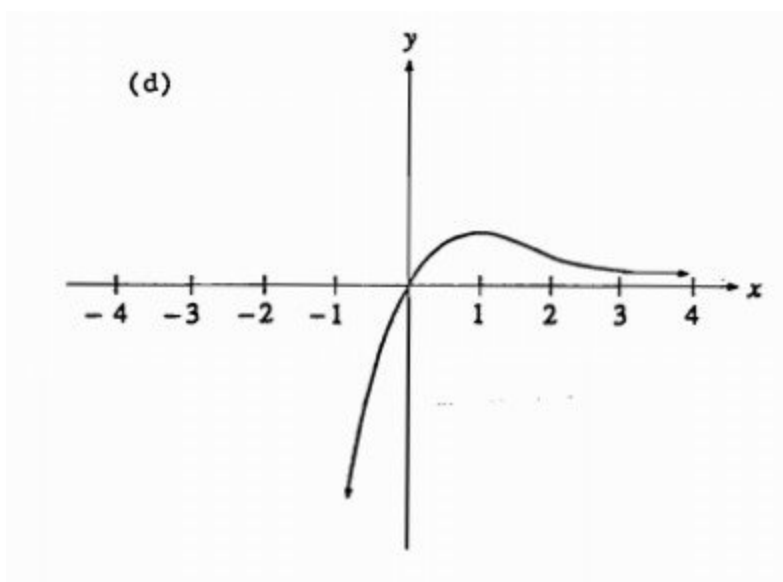
$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$

Part D

3 points are earned for the graph consistent with information found in (a), (b), and (c)



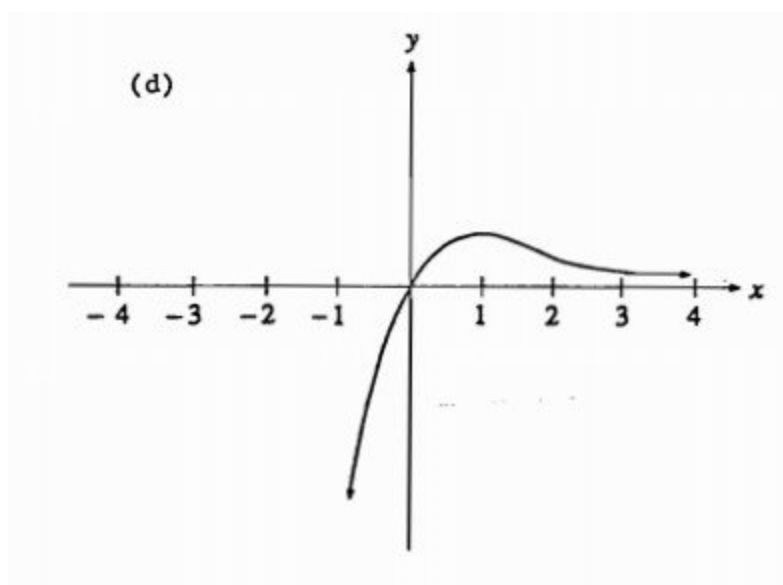
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✓

0	1	2	3
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The student response earns all of the following points:

3 points are earned for the graph consistent with information found in (a), (b), and (c)



4. Let F be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .
Write an equation of the horizontal asymptote for the graph of F .

Part A

1 point is earned for the correct equation

$$y = 0$$

✓

0	1
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The student response earns all of the following points:

1 point is earned for the correct equation

$$y = 0$$

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5. Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

(a) Find $\lim_{x \rightarrow \infty} g(x)$.

(b) For what positive value of b does g have an absolute maximum at $x = \frac{2}{3}$? Justify your answer.

(c) Find all values of b , if any, for which the graph of g has a point of inflection on the interval $0 < x < \infty$. Justify your answer.

Part A

1 point is earned for the answer

$$\lim_{x \rightarrow \infty} g(x) = 0$$



0

1

The student response earns all of the following points:

1 point is earned for the answer

$$\lim_{x \rightarrow \infty} g(x) = 0$$

Part B

2 points are earned for $g'(x)$

1 point is earned for solving $g' \left(\frac{2}{3} \right) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g' \left(\frac{2}{3} \right) = \left(\frac{1}{3} - b \right) e^{-2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = \left(\frac{2}{3} - x \right) e^{-x}.$$

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For $x < \frac{2}{3}$, $g'(x) > 0$ and for $x > \frac{2}{3}$, $g'(x) < 0$.

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum at $x = \frac{2}{3}$.



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for solving $g'\left(\frac{2}{3}\right) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g'\left(\frac{2}{3}\right) = \left(\frac{1}{3} - b\right)e^{-2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = \left(\frac{2}{3} - x\right)e^{-x}.$$

For $x < \frac{2}{3}$, $g'(x) > 0$ and for $x > \frac{2}{3}$, $g'(x) < 0$.

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum at $x = \frac{2}{3}$.

Part C

2 points are earned for $g''(x)$

1 point is earned for interval for b

1 point is earned for justification

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$$g''(x) = -e^{-x} - (1 - x - b)e^{-x} = (x - 2 + b)e^{-x}$$

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$.

Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for $g''(x)$

1 point is earned for interval for b

1 point is earned for justification

$$g''(x) = -e^{-x} - (1 - x - b)e^{-x} = (x - 2 + b)e^{-x}$$

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$.

Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.

6. Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

Find $\lim_{x \rightarrow \infty} g(x)$.

Part A

1 point is earned for the answer

$$\lim_{x \rightarrow \infty} g(x) = 0$$



0	1
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The student response earns all of the following points:

1 point is earned for the answer

$$\lim_{x \rightarrow \infty} g(x) = 0$$

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7. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be the function defined by $f(x) = \frac{ax^2+bx+2}{2x^2-8}$, where a and b are constants. The graph of f has a horizontal asymptote at $y = 3$, and f has a removable discontinuity at $x = 2$.

- Show that $a = 6$ and $b = -13$.
- To make f continuous at $x = 2$, $f(2)$ should be defined as what value? Justify your answer.
- At what value of x does f have a discontinuity due to a vertical asymptote? Give a reason for your answer.

Part A

The last point is earned for work that concludes with the correct values for both a, and b.

At most 3 out of 4 points if mathematical notation for limits is missing or incorrect.

Students do not have to show the factorization to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ recognizes $\lim_{x \rightarrow \infty} f(x) = 3$
- ☐ shows that $\lim_{x \rightarrow \infty} f(x) = \frac{a}{2}$
- ☐ sets $ax^2 + bx + 2 = 0$ at $x = 2$
- ☐ verification

Solution:

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$$\lim_{x \rightarrow \infty} f(x) = 3$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2 \left(a + \frac{b}{x} + \frac{2}{x^2} \right)}{x^2 \left(2 - \frac{8}{x^2} \right)} = \lim_{x \rightarrow \infty} \frac{a + \frac{b}{x} + \frac{2}{x^2}}{2 - \frac{8}{x^2}} = \frac{a}{2}$$

$$\Rightarrow a = 6$$

Since f has a removable discontinuity at $x = 2$ and the denominator $2x^2 - 8$ equals 0 at $x = 2$, the numerator must also equal 0 at $x = 2$.

$$0 = 6 \cdot 2^2 + 2b + 2 = 26 + 2b$$

$$\Rightarrow b = -13$$

Part B

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ uses $\lim_{x \rightarrow 2} f(x)$
- ☐ answer with justification

Solution:

f will be continuous at $x = 2$ if $f(2) = \lim_{x \rightarrow 2} f(x)$.

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{6x^2 - 13x + 2}{2x^2 - 8} = \lim_{x \rightarrow 2} \frac{(6x-1)(x-2)}{2(x+2)(x-2)} \\ &= \lim_{x \rightarrow 2} \frac{6x-1}{2(x+2)} = \frac{11}{8} \end{aligned}$$

$f(2)$ should be defined as $\frac{11}{8}$ to make f continuous at $x = 2$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response .

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0	1
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The student response accurately indicates $x = -2$ with reason.

Solution:

The graph of f has a vertical asymptote at $x = -2$ because the denominator $2x^2 - 8$ equals 0 at $x = -2$ but the numerator does not equal 0 at $x = -2$.

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8. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \frac{x^2+kx-2}{3x^2+4x+3} & \text{for } x < -2 \\ x^3 + 2 & \text{for } -2 \leq x < 0 \\ 0 & \text{for } x = 0 \\ \frac{2e^x}{2-e^x} & \text{for } x > 0 \end{cases}$$

Let f be the function defined above, where k is a constant.

- For what value of k , if any, is f continuous at $x = -2$? Justify your answer.
- What type of discontinuity does f have at $x = 0$? Give a reason for your answer.
- Find all horizontal asymptotes to the graph of f . Show the work that leads to your answer.

Part A

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $\lim_{x \rightarrow -2^-} f(x)$
- ☐ $\lim_{x \rightarrow -2^+} f(x)$
- ☐ justification equating limit with $f(-2)$

Solution:

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$$\begin{aligned}\lim_{x \rightarrow -2^-} f(x) &= \lim_{x \rightarrow -2^-} \frac{x^2 + kx - 2}{3x^2 + 4x + 3} = \frac{\lim_{x \rightarrow -2^-} (x^2 + kx - 2)}{\lim_{x \rightarrow -2^-} (3x^2 + 4x + 3)} \\ &= \frac{(-2)^2 + k(-2) - 2}{3(-2)^2 + 4(-2) + 3} = \frac{2 - 2k}{7}\end{aligned}$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (x^3 + 2) = (-2)^3 + 2 = -6$$

$$f(-2) = -6$$

$$f \text{ is continuous at } x = -2 \text{ if } \lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^+} f(x) = f(-2).$$

$$\frac{2 - 2k}{7} = -6 \Rightarrow 2 - 2k = -42 \Rightarrow k = 22$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

☐ $\lim_{x \rightarrow 0^-} f(x) \text{ and } \lim_{x \rightarrow 0^+} f(x)$

☐ answer
Solution:

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^3 + 2) = 2$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{2e^x}{2 - e^x} = \frac{2e^0}{2 - e^0} = 2$$

Because $\lim_{x \rightarrow 0} f(x) = 2$ and $f(0) = 0$, f has a removable discontinuity at $x = 0$.

Part C

At most 1 out of 2 points if both correct asymptotes are listed without reference to limits OR if one correct asymptote is listed without any incorrect asymptotes and/or reference to limits.

Students do not have to show the factorization to earn the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow \infty} f(x)$
- ☐ $\lim_{x \rightarrow -\infty} f(x)$

Solution:

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2e^x}{2 - e^x} = \lim_{x \rightarrow \infty} \frac{2e^x}{-e^x} = \lim_{x \rightarrow \infty} (-2) = -2$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{x^2 + kx - 2}{3x^2 + 4x + 3} = \lim_{x \rightarrow -\infty} \frac{x^2 \left(1 + \frac{k}{x} - \frac{2}{x^2}\right)}{x^2 \left(3 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{1 + \frac{k}{x} - \frac{2}{x^2}}{3 + \frac{4}{x} + \frac{3}{x^2}} = \frac{\lim_{x \rightarrow -\infty} \left(1 + \frac{k}{x} - \frac{2}{x^2}\right)}{\lim_{x \rightarrow -\infty} \left(3 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \frac{1 + 0 - 0}{3 + 0 + 0} = \frac{1}{3} \end{aligned}$$

The graph of f has horizontal asymptotes at $y = -2$ and $y = \frac{1}{3}$.

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9. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be the function defined by $f(x) = \frac{cx-5x^2}{2x^2+ax+b}$, where a , b , and c are constants. The graph of f has a vertical asymptote at $x = 1$, and f has a removable discontinuity at $x = -2$.

- Show that $a = 2$ and $b = -4$.
- Find the value of c . Justify your answer.
- To make f continuous at $x = -2$, $f(-2)$ should be defined as what value? Justify your answer.
- Write an equation for the horizontal asymptote to the graph of f . Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ sets $2x^2 + ax + b = 0$ for $x = -2$ and $x = 1$
- ☐ verification

Solution:

For f to have a removable discontinuity at $x = -2$ the denominator must equal zero at $x = -2$.

$$2(-2)^2 + a(-2) + b = 0 \Rightarrow 2a - b = 8$$

For f to have a vertical asymptote at $x = 1$ the denominator must equal zero at $x = 1$.

$$2(1)^2 + a(1) + b = 0 \Rightarrow a + b = -2$$

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$$2a - b = 8 \text{ and } a + b = -2 \Rightarrow 3a = 6 \Rightarrow a = 2$$

$$4 - b = 8 \Rightarrow b = 4 - 8 = -4$$

Part B

Work for the justification point may come before the answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ justification

Solution:

Since f has a removable discontinuity at $x = -2$ and the denominator equals 0 at $x = -2$, the numerator must also equal 0 at $x = -2$.

$$0 = c(-2) - 5(-2)^2 = -2c - 20$$

$$\Rightarrow c = \frac{20}{-2} = -10$$

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ uses $\lim_{x \rightarrow -2} f(x)$
- ☐ answer with justification

Solution:

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f will be continuous at $x = -2$ if $f(-2) = \lim_{x \rightarrow -2} f(x)$.

$$\begin{aligned}\lim_{x \rightarrow -2} f(x) &= \lim_{x \rightarrow -2} \frac{-10x - 5x^2}{2x^2 + 2x - 4} = \lim_{x \rightarrow -2} \frac{-5x(2+x)}{2(x-1)(x+2)} \\ &= \lim_{x \rightarrow -2} \frac{-5x}{2(x-1)} = \frac{-5(-2)}{2(-2-1)} = \frac{10}{-6} = -\frac{5}{3}\end{aligned}$$

$f(-2)$ should be defined as $-\frac{5}{3}$ to make f continuous at $x = -2$.

Part D

No point if correct asymptote is listed without reference to limits.

Students do not have to show the factorization to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
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The student response accurately includes an equation for the horizontal asymptote.

Solution:

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{-10x - 5x^2}{2x^2 + 2x - 4} = \lim_{x \rightarrow \infty} \frac{x^2(-\frac{10}{x} - 5)}{x^2(2 + \frac{2}{x} - \frac{4}{x^2})} \\ &= \lim_{x \rightarrow \infty} \frac{-\frac{10}{x} - 5}{2 + \frac{2}{x} - \frac{4}{x^2}} = \frac{\lim_{x \rightarrow \infty} (-\frac{10}{x} - 5)}{\lim_{x \rightarrow \infty} (2 + \frac{2}{x} - \frac{4}{x^2})} \\ &= \frac{0 - 5}{2 + 0 - 0} = -\frac{5}{2}\end{aligned}$$

The horizontal asymptote to the graph of f is $y = -\frac{5}{2}$.

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10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \frac{2x^2+5x-3}{x^2+4x+3} & \text{for } x < -3 \\ kx + \frac{1}{2} & \text{for } -3 \leq x \leq 0 \\ \frac{2^x}{3^{x-1}} & \text{for } x > 0 \end{cases}$$

Let f be the function defined above, where k is a constant.

- For what value of k , if any, is f continuous at $x = -3$? Justify your answer.
- What type of discontinuity does f have at $x = 0$? Give a reason for your answer.
- Find all horizontal asymptotes to the graph of f . Show the work that leads to your answer.

Part A

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{x \rightarrow -3^-} f(x)$
- ☐ $\lim_{x \rightarrow -3^+} f(x)$
- ☐ justification

Solution:

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$$\begin{aligned}
 \lim_{x \rightarrow -3^-} f(x) &= \lim_{x \rightarrow -3^-} \frac{2x^2 + 5x - 3}{x^2 + 4x + 3} = \lim_{x \rightarrow -3^-} \frac{(2x-1)(x+3)}{(x+1)(x+3)} \\
 &= \lim_{x \rightarrow -3^-} \frac{2x-1}{x+1} = \frac{\lim_{x \rightarrow -3^-} (2x-1)}{\lim_{x \rightarrow -3^-} (x+1)} \\
 &= \frac{2(-3)-1}{-3+1} = \frac{7}{2}
 \end{aligned}$$

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \left(kx + \frac{1}{2}\right) = -3k + \frac{1}{2}$$

$$f(-3) = -3k + \frac{1}{2}$$

$$f \text{ is continuous at } x = -3 \text{ if } \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^+} f(x) = f(-3).$$

$$-3k + \frac{1}{2} = \frac{7}{2} \Rightarrow -3k = 3 \Rightarrow k = -1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

he student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow 0^+} f(x)$
- ☐ answer

Solution:

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{2^x}{3^x - 1} = \infty$ because the numerator approaches 1, while the denominator approaches 0 from the right.

Therefore, f has a discontinuity due to a vertical asymptote at $x = 0$.

Part C

At most 1 out of 2 points if both correct asymptotes are listed without reference to limits OR if one correct asymptote is listed without any incorrect asymptotes and/or reference to limits.

Students do not have to show the factorization to earn the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow \infty} f(x)$
- ☐ $\lim_{x \rightarrow -\infty} f(x)$

Solution:

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2^x}{3^{x-1}} = \lim_{x \rightarrow \infty} \frac{2^x}{3^x} = \lim_{x \rightarrow \infty} \left(\frac{2}{3}\right)^x = 0 \text{ because } 0 < \frac{2}{3} < 1.$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{2x^2 + 5x - 3}{x^2 + 4x + 3} = \lim_{x \rightarrow -\infty} \frac{x^2 \left(2 + \frac{5}{x} - \frac{3}{x^2}\right)}{x^2 \left(1 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{2 + \frac{5}{x} - \frac{3}{x^2}}{1 + \frac{4}{x} + \frac{3}{x^2}} = \frac{\lim_{x \rightarrow -\infty} \left(2 + \frac{5}{x} - \frac{3}{x^2}\right)}{\lim_{x \rightarrow -\infty} \left(1 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \frac{2 + 0 - 0}{1 + 0 + 0} = 2 \end{aligned}$$

The graph of f has horizontal asymptotes at $y = 0$ and $y = 2$.

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11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The function f is given by $f(x) = \frac{x^3}{e^x}$.

- (a) Write an equation for the line tangent to the graph of f at $x = 5$.
- (b) Write an equation that involves a limit and demonstrates that the x -axis is an asymptote for the graph of f .
- (c) Find the x -coordinate of each critical point of f . Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.
- (d) Find the average value of f' , the derivative of f , on the interval $[-1, 5]$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$f'(x) = \frac{3x^2 - x^3}{e^x}$ earns the first 2 points.

An unlabeled correct $f'(x)$ earns the first 2 points.

To earn the first point:

- If the quotient rule was used, the presented derivative must include a sum or difference, a factor of e^x , and a polynomial in each term of the numerator.
- If the product rule was used, the presented derivative must include a factor of e^{-x} in each term.

A response that includes $f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{e^{x^2}}$ earns the first point, does not earn the second point, and is eligible for the third point. In this case the response must correctly evaluate this expression at $x = 5$, and use the resulting value as the slope of the tangent line.

The second point cannot be earned without the first. The second point is earned for a correct expression for $f'(x)$ or a correct value of $f'(5)$.

A response that labels the values $f(5)$ and $f'(5)$ correctly with no other supporting work does not earn the first 2 points but is eligible for the third point.

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A response that presents a correct $f'(x)$ but an incorrect $f'(5)$ earns the first 2 points, but does not earn the third point.

A response that presents a correct equation of the tangent line with no supporting work earns no points in this part.

A response with any arithmetic error in $f'(x)$ does not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Form of quotient rule (or product rule)
- ☐ $f'(x)$ or $f'(5)$
- ☐ Answer

Solution:

$$f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{(e^x)^2} = \frac{3x^2 - x^3}{e^x}$$

$$f'(5) = \frac{75 - 125}{e^5} = \frac{-50}{e^5}$$

$$f(5) = \frac{125}{e^5}$$

An equation for the tangent line is $y = \frac{125}{e^5} - \frac{50}{e^5}(x - 5)$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$\lim_{x \rightarrow \infty} f(x) = 0$ or $\lim_{x \rightarrow \infty} \frac{x^3}{e^x} = 0$ earns the point.

A response that presents an expression, rather than an equation, does not earn the point.

A response that presents any additional incorrect limits does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

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· Equation

Solution:

$$\lim_{x \rightarrow \infty} f(x) = 0$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$(3x^2 - x^3) = 0$ earns the first point.

A response that presents only one of $x = 0$ or $x = 3$ along with correct answer and justification earns 1 of the last 2 points.

A response that lists additional values of x as critical points does not earn either of the last 2 points.

The second derivative may be used to justify the relative maximum at $x = 3$. Since $f''(x) = 0$ at $x = 0$, it is not sufficient to use the second derivative to conclude that $x = 0$ is neither a maximum or minimum value; more justification is needed.

A sign chart with no explanation is not sufficient justification.

Because the interval is not closed, the Extreme Value Theorem may not be used.

A response that imports e^{x^2} in the denominator of the derivative from part (a) is eligible for all 3 points.

A response that imports either $f'(x) = \frac{e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ or $f'(x) = \frac{-e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ from part (a) is eligible for all 3 points.

· An import of $f'(x) = \frac{e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ should find critical points $x = -3$ and $x = 0$ AND conclude that f has a relative minimum at $x = -3$ and neither a minimum nor a maximum at $x = 0$.

· An import of $f'(x) = \frac{-e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ should find critical points $x = 0$ and $x = 3$ AND conclude that f has neither a minimum nor a maximum at $x = 0$ and a relative minimum at $x = 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $f'(x) = 0$
 $x = 0$ and $x = 3$ only

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☐☐

Answer with justification

Solution:

$$f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{(e^x)^2} = \frac{x^2 e^x (3 - x)}{e^{2x}} = 0 \Rightarrow x = 0, x = 3$$

$f'(x)$ does not change sign at $x = 0$. Therefore, f has neither a minimum nor a maximum at $x = 0$.

$f'(x)$ changes sign from positive to negative at $x = 3$. Therefore, f has a relative maximum at $x = 3$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A factor of $\frac{1}{6}$ is not needed to earn the first point.

The definite integral can be presented in a single or multiple steps to earn the first point. For example, the following response earns both points: $\int f'(x) dx = f(x)$, so the average value is $\frac{1}{6} \left(\frac{125}{e^5} + \frac{1}{e^{-1}} \right)$.

$f(5) - f(-1)$ earns the first point and $\frac{1}{6} \left(\frac{125}{e^5} - \left(\frac{-1}{e^{-1}} \right) \right)$ earns both points, provided there are no subsequent errors in simplification.

A response that presents a correct integral along with the correct average value, but provides incorrect or incomplete communication, earns 1 out of 2 points. For example, the following response earns 1 out of 2 points:

$$\int_{-1}^5 f'(x) dx = \left(\frac{125}{e^5} + e \right) = \frac{1}{6} \left(\frac{125}{e^5} + e \right).$$

$\int_{-1}^5 f'(x) dx$ with an incorrect $f'(x)$ imported from part (a) or part (c) earns the first point.

A response with incorrect limits earns no points.

A response of $\frac{1}{6} \cdot \frac{125}{e^5} - \frac{-1}{e^{-1}}$ or $\frac{1}{6} \cdot f(5) - f(1)$ earns the first point but not the second point.

A response that uses an incorrect value of $f(5)$ or $f(-1)$ does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 15☐ Integral☐ Answer**Solution:**

$$\frac{1}{5 - (-1)} \int_{-1}^5 f'(x) \, dx$$

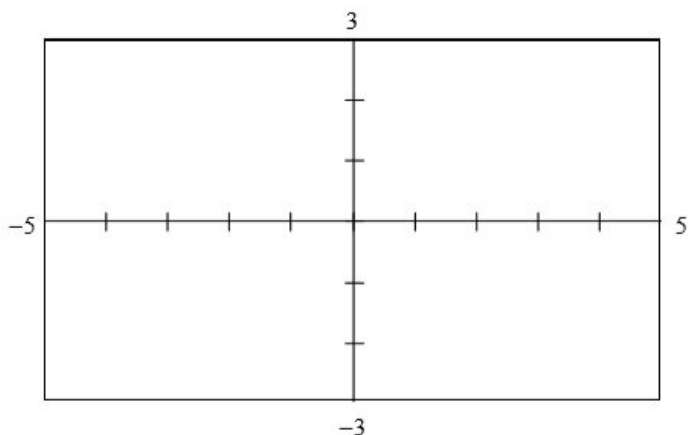
$$= \frac{1}{6} (f(5) - f(-1)) = \frac{1}{6} \left(\frac{125}{e^5} - \left(\frac{-1}{e^{-1}} \right) \right) = \frac{1}{6} \left(\frac{125}{e^5} + e \right)$$

Unit 1 AP Topics 15

12. Let f be the function given by $f(x) = \frac{2x}{\sqrt{x^2+x+1}}$.

(a) Find the domain of f . Justify your answer.

(b) In the viewing window provided below, sketch the graph of f .



Viewing Window
 $[-5, 5] \times [-3, 3]$

(c) Write an equation for each horizontal asymptote of the graph of f .

(d) Find the range of f . Use $f'(x)$ to justify your answer.

Note: $f'(x) = x + 2 / (x^2 + x + 1)^{3/2}$

Part A

1 point is earned for the correct answer

1 point is earned for the correct justification

Domain: all real numbers since $x^2 + x + 1 > 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer

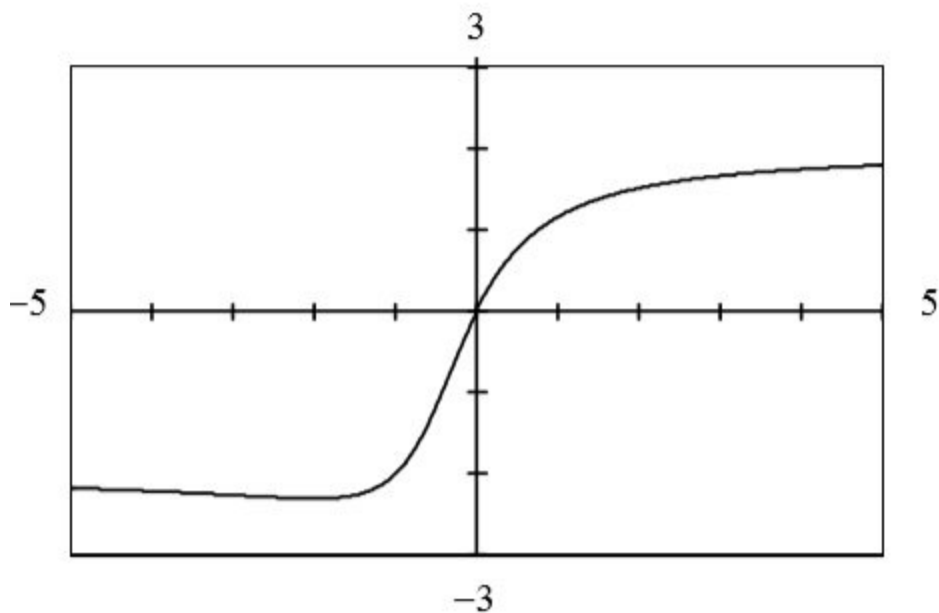
1 point is earned for the correct justification

Domain: all real numbers since $x^2 + x + 1 > 0$.

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Part B

1 point is earned for the correct graph



Viewing Window
 $[-5, 5] \times [-3, 3]$



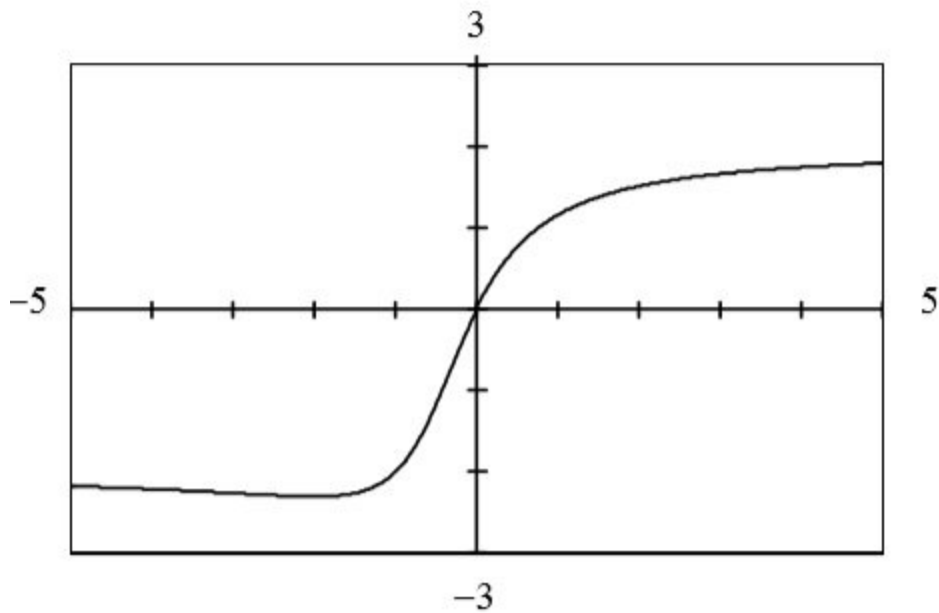
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1

The student response earns all of the following points:

1 point is earned for the correct graph

Unit 1 AP Topics 15



Viewing Window
 $[-5, 5] \times [-3, 3]$

Part C

1 point is earned for the correct equation $y = 2$

$$y = 2$$

1 point is earned for the correct equation $y = -2$

$$y = -2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct equation $y = 2$

$$y = 2$$

1 point is earned for the correct equation $y = -2$

$$y = -2$$

Part D

Unit 1 AP Topics 15

1 point is earned for the correct right endpoint

1 point is earned for the correct left endpoint (min)

1 point is earned for the correct inclusion/exclusion of endpoints

Range is $\left[\frac{-4}{\sqrt{3}}, 2\right)$ or $-2.309 \leq y < 2$.

1 point is earned for using sign of f' to justify

Sign of f' is given by $\frac{-}{+}$ $\frac{-}{-2}$, implying that f has a minimum at $x = -2$ and, with asymptotes,

that f is never greater than 2.

Note: max 2/4 (1 - 1 - 0 - 0) if using incorrect asymptotes



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct right endpoint

1 point is earned for the correct left endpoint (min)

1 point is earned for the correct inclusion/exclusion of endpoints

Range is $\left[\frac{-4}{\sqrt{3}}, 2\right)$ or $-2.309 \leq y < 2$.

1 point is earned for using sign of f' to justify

Sign of f' is given by $\frac{-}{+}$ $\frac{-}{-2}$, implying that f has a minimum at $x = -2$ and, with asymptotes,

that f is never greater than 2.

Note: max 2/4 (1 - 1 - 0 - 0) if using incorrect asymptotes

Unit 1 AP Topics 15

13. Let f be the function given by $f(x) = \frac{2x}{\sqrt{x^2+x+1}}$.

Write an equation for each horizontal asymptote of the graph of f .

Part C

1 point is earned for the correct equation $y = 2$

$$y = 2$$

1 point is earned for the correct equation $y = -2$

$$y = -2$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct equation $y = 2$

$$y = 2$$

1 point is earned for the correct equation $y = -2$

$$y = -2$$

14. For $x \geq 0$, the horizontal line $y = 2$ is an asymptote for the graph of the function f . Which of the following statements must be true?

(A) $f(0) = 2$

(B) $f(x) \neq 2$ for all $x \geq 0$

(C) $f(2)$ is undefined.

(D) $\lim_{x \rightarrow 2} f(x) = \infty$

(E) $\lim_{x \rightarrow \infty} f(x) = 2$



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15. For which of the following does $\lim_{x \rightarrow \infty} f(x) = 0$?

I. $f(x) = \frac{\ln x}{x^{99}}$

II. $f(x) = \frac{e^x}{\ln x}$

III. $f(x) = \frac{x^{99}}{e^x}$

(A) I only

(B) II only

(C) III only

(D) I and II only

(E) I and III only



Unit 1 AP Topics 15

16. Let f be the function that is given by $f(x) = ax + b/x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y -axis.

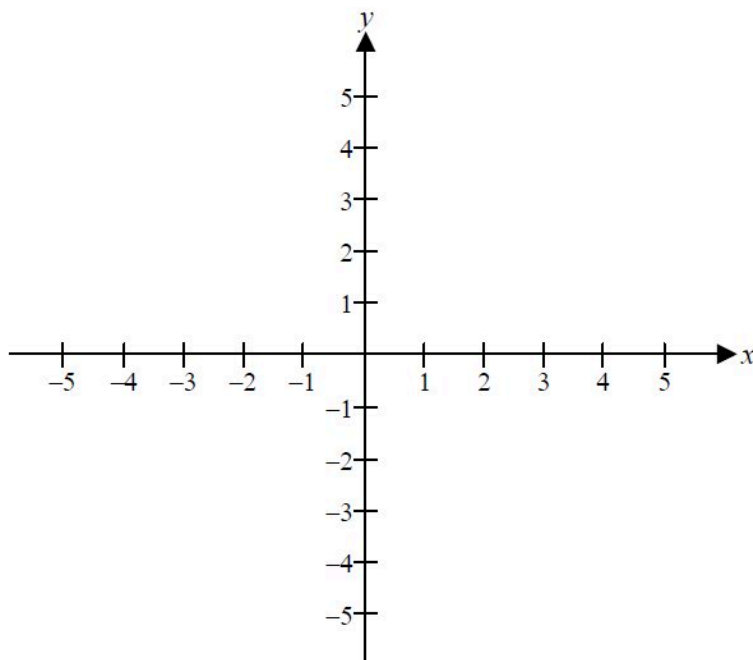
(ii) $\lim_{x \rightarrow 2^+} f(x) = +\infty$

(iii) $f'(1) = -2$

(a) Determine the values of a , b , and c .

(b) Write an equation for each vertical and each horizontal asymptote of the graph of f .

(c) Sketch the graph of f in the xy -plane provided below.

**Part A**

1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y -axis then f is even

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$$f(-x) = f(x) \quad \therefore \boxed{a = 0}$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \quad \therefore \boxed{c = 4}$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \quad \therefore \boxed{b = 9}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f'(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y -axis then f is even

$$f(-x) = f(x) \quad \therefore \boxed{a = 0}$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \quad \therefore \boxed{c = 4}$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \quad \therefore \boxed{b = 9}$$

Part B

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

Unit 1 AP Topics 15VERTICAL: $x = 2$, $x = -2$ HORIZONTAL: $y = 0$ 

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

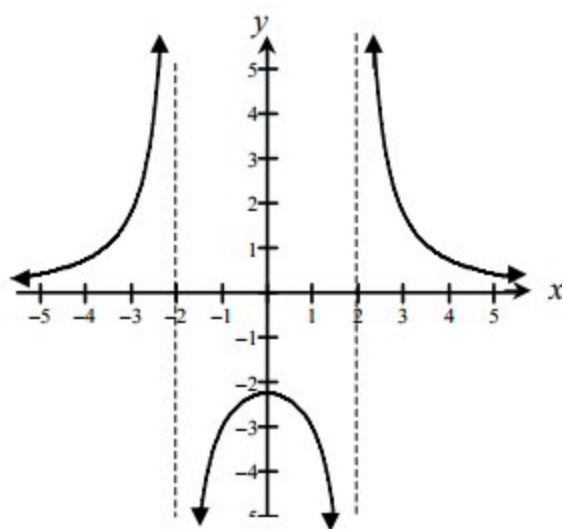
Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

VERTICAL: $x = 2$, $x = -2$ HORIZONTAL: $y = 0$ **Part C**

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y -int

Unit 1 AP Topics 15



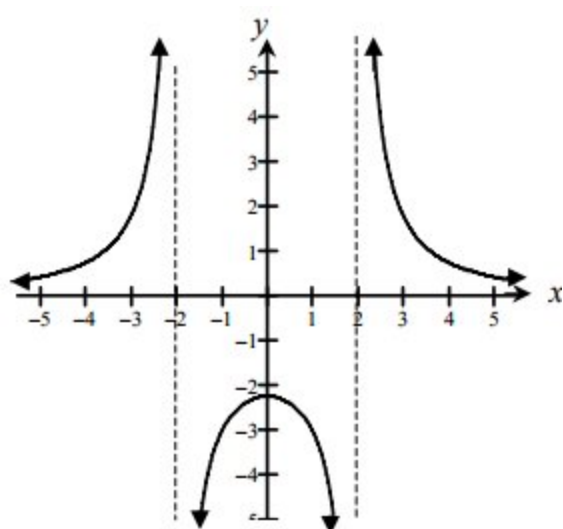
0	1	2	3
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The student response earns all of the following points:

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y -int



Unit 1 AP Topics 15

17. Let f be the function that is given by $f(x) = ax + b/x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y -axis.

(ii) $\lim_{x \rightarrow 2^+} f(x) = +\infty$

(iii) $f'(1) = -2$

Write an equation for each vertical and each horizontal asymptote of the graph of f .

Part B

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

Vertical : $x = 2, x = -2$

Horizontal : $y = 0$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

Vertical : $x = 2, x = -2$

Horizontal : $y = 0$

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18. Let f be the function defined by $f(x) = xe^{1-x}$ for all real numbers x .
Find the range of f .

Part B

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range : $(-\infty, 1]$



0	1	2
---	---	---

The student response earns none of the following points:

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

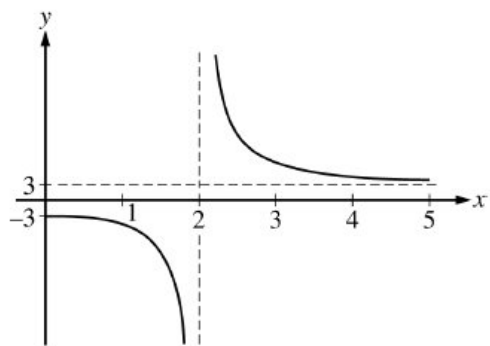
Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range : $(-\infty, 1]$

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19.



The function f is given by $f(x) = \frac{ax^2 + 12}{x^2 + b}$. The figure above shows a portion of the graph of f . Which of the following could be the values of the constants a and b ?

(A) $a = -3, b = 2$

(B) $a = 2, b = -3$

(C) $a = 2, b = -2$

(D) $a = 3, b = -4$

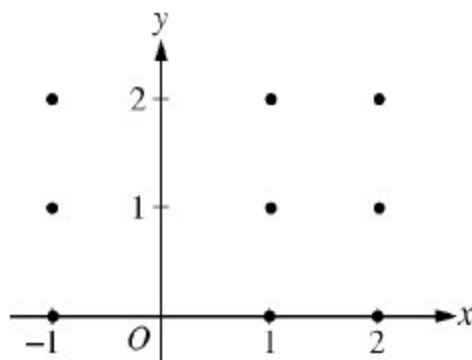
(E) $a = 3, b = 4$



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20. Consider the differential equation $\frac{dy}{dx} = \frac{y-1}{x^2}$, where $x \neq 0$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



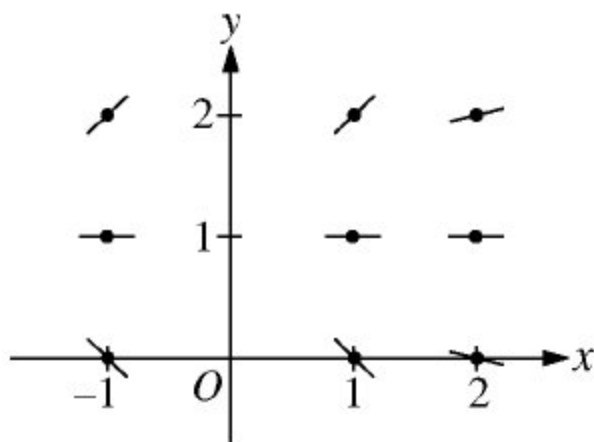
(b) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(2) = 0$.

(c) For the particular solution $y = f(x)$ described in part (b), find $\lim_{x \rightarrow \infty} f(x)$.

Part A

1 point: For zero slopes

1 point: For all other slopes



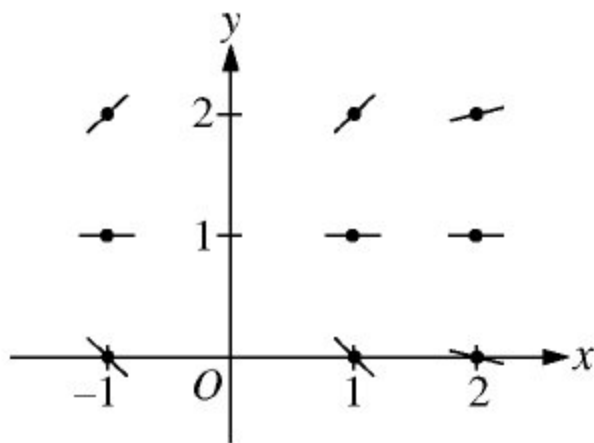
0	1	2
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The response earns all of the following points:

1 point: For zero slopes

1 point: For all other slopes

**Part B**

The response can earn up to 6 points:

Note: 0/6 if no separation of variables

Note: max 3/6 if no constant of integration

1 point: For separation of variables

2 points: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

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$$\frac{1}{y-1} dy = \frac{1}{x^2} dx$$

$$\ln |y - 1| = -\frac{1}{x} + C$$

$$|y - 1| = e^{-\frac{1}{x} + C}$$

$$|y - 1| = e^C e^{-\frac{1}{x}}$$

$$y - 1 = k e^{-\frac{1}{x}}, \text{ where } k = \pm e^C$$

$$-1 = k e^{-\frac{1}{2}}$$

$$k = -e^{\frac{1}{2}}$$

$$f(x) = 1 - e^{(\frac{1}{2} - \frac{1}{x})}, x > 0$$



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The response earns all of the following points:

Note: 0/6 if no separation of variables

Note: max 3/6 if no constant of integration

1 point: For separation of variables

2 points: For the antiderivatives

1 point: For the constant of integration

1 point: For use of the initial condition

1 point: For the solution of y

$$\frac{1}{y-1} dy = \frac{1}{x^2} dx$$

$$\ln |y - 1| = -\frac{1}{x} + C$$

$$|y - 1| = e^{-\frac{1}{x} + C}$$

$$|y - 1| = e^C e^{-\frac{1}{x}}$$

$$y - 1 = k e^{-\frac{1}{x}}, \text{ where } k = \pm e^C$$

$$-1 = k e^{-\frac{1}{2}}$$

$$k = -e^{\frac{1}{2}}$$

$$f(x) = 1 - e^{(\frac{1}{2} - \frac{1}{x})}, x > 0$$

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Part C

The response can earn up to 1 point:

1 point: For the correct limit

$$\lim_{x \rightarrow \infty} 1 - e^{\left(\frac{1}{2} - \frac{1}{x}\right)} = 1 - \sqrt{e}$$



0	1
---	---

The response earns the following point:

1 point: For the correct limit

$$\lim_{x \rightarrow \infty} 1 - e^{\left(\frac{1}{2} - \frac{1}{x}\right)} = 1 - \sqrt{e}$$

21. Consider the differential equation $\frac{dy}{dx} = \frac{y-1}{x^2}$, where $x \neq 0$.

For the particular solution $y = f(x)$ described in the previous question, find $\lim_{x \rightarrow \infty} f(x)$.

Part C

The response can earn up to 1 point:

1 point: For the correct limit

$$\lim_{x \rightarrow \infty} 1 - e^{\left(\frac{1}{2} - \frac{1}{x}\right)} = 1 - \sqrt{e}$$



0	1
---	---

The response earns the following point:

1 point: For the correct limit

$$\lim_{x \rightarrow \infty} 1 - e^{\left(\frac{1}{2} - \frac{1}{x}\right)} = 1 - \sqrt{e}$$

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22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = 10 - 2y$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 2$.

- Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 0$. Use the tangent line to approximate $f(0.5)$.
- Find the value of $\frac{d^2y}{dx^2}$ at the point $(0, 2)$. Is the graph of $y = f(x)$ concave up or concave down at the point $(0, 2)$? Give a reason for your answer.
- Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0) = 2$.
- For the particular solution $y = f(x)$ found in part (c), find $\lim_{x \rightarrow \infty} f(x)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0,2)} = 10 - 2 \cdot 2 = 6$$

An equation for the line tangent to the graph of $y = f(x)$ at $x = 0$ is $y = 6x + 2$.

$$f(0.5) \approx 6 \cdot 0.5 + 2 = 5$$

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Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{d^2y}{dx^2} \Big|_{(x,y)=(0,2)}$
- ☐ concave down with reason

Solution:

$$\frac{d^2y}{dx^2} = -2 \frac{dy}{dx} = -2(10 - 2y) = 4y - 20$$

$$\frac{d^2y}{dx^2} \Big|_{(x,y)=(0,2)} = 4 \cdot 2 - 20 = -12$$

Because $\frac{d^2y}{dx^2} \Big|_{(x,y)=(0,2)} < 0$ and $\frac{d^2y}{dx^2}$ is continuous, the graph of $y = f(x)$ is concave down at the point $(0, 2)$.

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

The fourth point requires an expression for y .

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ separation of variables
- ☐ antiderivative
- ☐ constant of integration and uses initial condition

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☐ solves for y

Solution:

$$\frac{1}{10-2y} dy = dx$$

$$\int \frac{1}{10-2y} dy = \int 1 dx$$

$$-\frac{1}{2} \ln |10-2y| = x + C$$

Since the solution curve includes the point $(0, 2)$,

$$-\frac{1}{2} \ln (10-2y) = x + C$$

$$-\frac{1}{2} \ln (10-2 \cdot 2) = 0 + C \Rightarrow C = -\frac{1}{2} \ln 6$$

$$-\frac{1}{2} \ln (10-2y) = x - \frac{1}{2} \ln 6$$

$$\ln (10-2y) = -2x + \ln 6$$

$$10-2y = e^{-2x+\ln 6} = 6e^{-2x}$$

$$y = 5 - 3e^{-2x}$$

Part D

No supporting work is required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\lim_{x \rightarrow \infty} (5 - 3e^{-2x}) = 5 - 3 \cdot 0 = 5$$

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23. The line $y = 5$ is a horizontal asymptote to the graph of which of the following functions?

(A) $y = \frac{\sin(5x)}{x}$

(B) $y = 5x$

(C) $y = \frac{1}{x-5}$

(D) $y = \frac{5x}{1-x}$

(E) $y = \frac{20x^2 - x}{1 + 4x^2}$ ✓

24. What is $\lim_{x \rightarrow \infty} \frac{x^2 - 4}{2 + x - 4x^2}$?

(A) -2

(B) $-\frac{1}{4}$ ✓

(C) $\frac{1}{2}$

(D) 1

(E) The limit does not exist.

25. $\lim_{x \rightarrow \infty} \frac{x^3}{e^{3x}}$ is

(A) 0 ✓

(B) $\frac{2}{9}$

(C) $\frac{2}{3}$

(D) 1

(E) infinite

26. The continuous function f is positive and has domain $x > 0$. If the asymptotes of the graph of f are $x = 0$ and $y = 2$, which of the following statements must be true?

(A) $\lim_{x \rightarrow 0^+} f(x) = \infty$ and $\lim_{x \rightarrow 2} f(x) = \infty$

(B) $\lim_{x \rightarrow 0^+} f(x) = 2$ and $\lim_{x \rightarrow \infty} f(x) = 0$

(C) $\lim_{x \rightarrow 0^+} f(x) = \infty$ and $\lim_{x \rightarrow \infty} f(x) = 2$ ✓

(D) $\lim_{x \rightarrow 2} f(x) = \infty$ and $\lim_{x \rightarrow \infty} f(x) = 2$

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27. The graph of which of the following functions has exactly one horizontal asymptote and no vertical asymptotes?

(A) $y = \frac{1}{x^2+1}$



(B) $y = \frac{1}{x^3+1}$

(C) $y = \frac{1}{e^x-1}$

(D) $y = \frac{1}{e^x+1}$

28. What are the equations of the horizontal asymptotes of the graph of $y = \frac{2x}{\sqrt{x^2-1}}$?

(A) $y = 0$ only

(B) $y = 1$ only

(C) $y = 2$ only

(D) $y = -2$ only and $y = 2$ only



(E) $y = -1$ only and $y = 1$ only

29. Let f be the function given by $f(x) = \frac{(x-4)(2x-3)}{(x-1)^2}$. If the line $y = b$ is a horizontal asymptote to the graph of f , then $b =$

(A) 0

(B) 1

(C) 2



(D) 3

(E) 4

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30. The line $y = 2$ is a horizontal asymptote to the graph of which of the following functions?

(A) $y = \frac{x+2\sin x}{x-2}$

(B) $y = \frac{2x^2+\sin x}{x^2}$ ✓

(C) $y = \frac{2x^2+\sin x}{4-x^2}$

(D) $y = \frac{2^x+2\sin x}{2^x}$

Answer B

Correct.

$$\lim_{x \rightarrow \infty} \frac{2x^2 + \sin x}{x^2} = \lim_{x \rightarrow \infty} \frac{x^2 \left(2 + \frac{\sin x}{x^2} \right)}{x^2} = \lim_{x \rightarrow \infty} \left(2 + \frac{\sin x}{x^2} \right) = 2 + 0 = 2$$

$$\lim_{x \rightarrow -\infty} \frac{2x^2 + \sin x}{x^2} = \lim_{x \rightarrow -\infty} \frac{x^2 \left(2 + \frac{\sin x}{x^2} \right)}{x^2} = \lim_{x \rightarrow -\infty} \left(2 + \frac{\sin x}{x^2} \right) = 2 + 0 = 2$$

Therefore the line $y = 2$ is a horizontal asymptote for $y = \frac{2x^2 + \sin x}{x^2}$ as x approaches both positive infinity and negative infinity.

31. The function f is given by $f(x) = \sin\left(\frac{x+1}{x^2}\right)$. Which of the following statements are true?

I. The graph of f has a horizontal asymptote at $y=0$.

II. The graph of f has a horizontal asymptote at $y=1$.

III. The graph of f has a vertical asymptote at $x=0$.

(A) I only ✓

(B) II only

(C) III only

(D) I and III only

(E) II and III only

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32. Let f be the function defined by $f(x) = \frac{(3x+8)(5-4x)}{(2x+1)^2}$. Which of the following is a horizontal asymptote to the graph of f ?

(A) $y = -6$

(B) $y = -3$ ✓

(C) $y = -\frac{1}{2}$

(D) $y = 0$

(E) $y = \frac{3}{2}$

33. The graph of which of the following equations has $y = 1$ as an asymptote?

(A) $y = \ln x$

(B) $y = \sin x$

(C) $y = \frac{x}{x+1}$ ✓

(D) $y = \frac{x^2}{x-1}$

(E) $y = e^{-x}$

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34. Let f be the function given by $f(x) = \frac{x}{\sqrt{x^2 - 4}}$.

- (a) Find the domain of f .
- (b) Write an equation for each vertical asymptote to the graph of f .
- (c) Write an equation for each horizontal asymptote to the graph of f .
- (d) Find $f'(x)$.

Part A

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$$x < -2 \text{ or } x > 2$$

$$\text{OR } |x| > 2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$$x < -2 \text{ or } x > 2$$

$$\text{OR } |x| > 2$$

Part B

1 point is earned for finding value of x

$$x = 2, x = -2$$

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0	1
---	---

The student response earns all of the following points:

1 point is earned for finding value of x

$$x = 2, x = -2$$

Part C

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x \rightarrow +\infty} \frac{x}{\sqrt{x^2 - 4}} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{x^2 - 4}} = -1$$

$$y = 1, y = -1$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x \rightarrow +\infty} \frac{x}{\sqrt{x^2 - 4}} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{x^2 - 4}} = -1$$

$$y = 1, y = -1$$

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Part D

3 points are earned for finding $f'(x)$

2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2-4} - x \left[\frac{1}{2}(x^2-4)^{-1/2} 2x \right]}{x^2-4} \\ &= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\ &= \frac{-4}{(x^2-4)^{3/2}} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

3 points are earned for finding $f'(x)$

2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2-4} - x \left[\frac{1}{2}(x^2-4)^{-1/2} 2x \right]}{x^2-4} \\ &= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\ &= \frac{-4}{(x^2-4)^{3/2}} \end{aligned}$$

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35. $\lim_{n \rightarrow \infty} \frac{3n^3 - 5n}{n^3 - 2n^2 + 1}$ is

- (A) -5
(B) -2
(C) 1

(D) 3

(E) nonexistent



36. $\lim_{x \rightarrow \infty} \frac{(2x-1)(3-x)}{(x-1)(x+3)}$ is

(A) -3

(B) -2

(C) 2

(D) 3

(E) nonexistent



37. $\lim_{x \rightarrow \infty} \frac{x^3 - 2x^2 + 3x - 4}{4x^3 - 3x^2 + 2x - 1} =$

(A) 4

(B) 1

(C) 1/4

(D) 0

(E) -1



38. If the graph of $y = \frac{ax+b}{x+c}$ has a horizontal asymptote $y=2$ and a vertical asymptote $x=-3$, then $a+c=$

(A) -5

(B) -1

(C) 0

(D) 1

(E) 5



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39. Write an equation for each horizontal asymptote to the graph of f .

Part C

1 point is earned for analysis in negative direction

1 point is earned for analysis in positive direction

1 point is earned for solution

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 - 4}} = 1$$

$$\lim_{x \rightarrow \infty} \frac{x}{x^2 - 4} = -1$$

$$y = 1, y = -1$$



0	1	2	3
---	---	---	---

The student response earns none of the following points:

1 point is earned for analysis in negative direction

1 point is earned for analysis in positive direction

1 point is earned for solution

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 - 4}} = 1$$

$$\lim_{x \rightarrow \infty} \frac{x}{x^2 - 4} = -1$$

$$y = 1, y = -1$$

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40. Let f be the function given by $f(x) = \frac{\sqrt{abs}x-2}{x-2}$.

Find the range of f .

Part D

1 point is earned for answer

$$-1 < y \leq 1$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for answer

$$-1 < y \leq 1$$

41. $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^4+1}}{4x^2+3}$

(A) 1/3

(B) 3/4

(C) 3/2

(D) 9/4

(E) infinite



42. For which of the following pairs of functions f and g is $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ infinite?

(A) $f(x) = x^2 + 2x$ and $g(x) = x^2 + \ln x$

(B) $f(x) = 3x^3$ and $g(x) = x^4$

(C) $f(x) = 3^x$ and $g(x) = x^3$

(D) $f(x) = 3e^x + x^3$ and $g(x) = 2e^x + x^2$

(E) $f(x) = \ln(3x)$ and $g(x) = \ln(2x)$



43. $\lim_{x \rightarrow -\infty} \frac{3+2^x}{4-5^x}$ is

(A) $-2/5$

(B) 0

(C) 3/4

(D) non existent



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44. Let $P(t)$ represent the number of wolves in a population at time t years, when $t \geq 0$. The population $P(t)$ is increasing at a rate directly proportional to $800 - P(t)$, where the constant of proportionality is k .
- (a) If $P(0) = 500$, find $P(t)$ in terms of t and k .
- (b) If $P(2) = 700$, find k .
- (c) Find $\lim_{t \rightarrow \infty} P(t)$.

Part A

1 point is earned for differential equation

1 point is earned if separates variables

2 points are earned for anti differentiation

1 point is **deducted** for errors

1 points is earned for solving for constant

1 point is earned if solves for $p(t)$

Note: If no differential equation solved, then not eligible for last two points

$$P'(t) = k(800 - P(t))$$

$$\frac{dP}{800 - P} = k dt$$

$$-\ln|800 - P| = kt + C_0$$

$$|800 - P| = C_1 e^{-kt}$$

$$800 - 500 = C_1 e^0$$

$$C_1 = 300$$

$$\therefore P(t) = 800 - 300e^{-kt}$$



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for differential equation

1 point is earned if separates variables

2 points are earned for anti differentiation

1 point is **deducted** for errors

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1 point is earned for solving for constant

1 point is earned if solves for $p(t)$

Note: If no differential equation solved, then not eligible for last two points

$$P'(t) = k(800 - P(t))$$

$$\frac{dP}{800 - P} = k dt$$

$$-\ln|800 - P| = kt + C_0$$

$$|800 - P| = C_1 e^{-kt}$$

$$800 - 500 = C_1 e^0$$

$$C_1 = 300$$

$$\therefore P(t) = 800 - 300e^{-kt}$$

Part B

1 point is earned for setting $P(2)$ equal to 700

1 point is earned for solving for k

$$P(2) = 700 = 800 - 300e^{-2k}$$

$$k = \frac{\ln 3}{2} \approx 0.549$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for setting $P(2)$ equal to 700

1 point is earned for solving for k

$$P(2) = 700 = 800 - 300e^{-2k}$$

$$k = \frac{\ln 3}{2} \approx 0.549$$

Part C

1 point is earned for limits

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$$\lim_{t \rightarrow \infty} \left(800 - 300e^{-\frac{\ln 3}{2}t} \right) = 800$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for limits

$$\lim_{t \rightarrow \infty} \left(800 - 300e^{-\frac{\ln 3}{2}t} \right) = 800$$

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45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$F(t) = \begin{cases} 2^{t+1} - 2 & \text{for } 0 \leq t < 7 \\ \frac{800t - 4800}{t - 4} & \text{for } t \geq 7 \end{cases}$$

The amount of money raised during a fund-raising campaign is modeled by the function F defined above, where $F(t)$ is measured in United States dollars and t is the time in days since the campaign began.

- Find $\lim_{t \rightarrow \infty} F(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} F(t)$ in the context of the problem.
- Is the function F continuous at $t = 7$? Justify your answer.
- The amount of money raised during a competing fund-raising campaign is modeled by the function M defined by $M(t) = \frac{240(2^t - 1)}{(2^t + 36)}$, where $M(t)$ is measured in United States dollars and t is the time in days since that campaign began. According to this model, is there a time t , for $0 \leq t \leq 2$, at which the amount of money raised is 10 dollars? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the amount raised.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} F(t)$
- ☐ interpretation

Solution:

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$$\lim_{t \rightarrow \infty} F(t) = \lim_{t \rightarrow \infty} \frac{800t - 4800}{t - 4} = \frac{800}{1} = 800$$

As time increases, the amount raised by the fund-raising campaign approaches \$800.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 7^-} F(t)$
- ☐ $\lim_{t \rightarrow 8^+} F(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 7^-} F(t) = \lim_{t \rightarrow 7^-} (2^{t+1} - 2) = 2^8 - 2 = 254$$

$$\lim_{t \rightarrow 7^+} F(t) = \lim_{t \rightarrow 7^+} \frac{800t - 4800}{t - 4} = \frac{800 \cdot 7 - 4800}{7 - 4} = \frac{800}{3}$$

No, F is not continuous at $t = 7$ because $\lim_{t \rightarrow 7^-} F(t) \neq \lim_{t \rightarrow 7^+} F(t)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

- ☐ answer with justification
- ☐ specific reference to IVT

Unit 1 AP Topics 15**Solution:**

M is continuous for $0 \leq t \leq 2$.

$$M(0) = \frac{240(2^0 - 1)}{2^0 + 36} = \frac{240(1 - 1)}{1 + 36} = 0$$

$$M(2) = \frac{240(2^2 - 1)}{2^2 + 36} = \frac{240 \cdot 3}{40} = 18$$

$$M(0) = 0 < 10 < 18 = M(2)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 2$, such that $M(t) = 10$.

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46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$Y(t) = \begin{cases} 10e^{0.05t} & \text{for } 0 \leq t \leq 10 \\ f(t) & \text{for } 10 < t \leq 12 \\ \frac{600}{20+10e^{-0.05(t-12)}} & \text{for } t > 12 \end{cases}$$

The mass of yeast in a culture, measured in milligrams, at time t hours is modeled by the function Y defined above, where f is a continuous function such that $f(12) = 20$.

Find $\lim_{t \rightarrow \infty} Y(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} Y(t)$ in the context of the problem.

Is the function Y continuous at $t = 12$? Justify your answer.

The function Y is continuous at $t = 10$. Is there a time t , for $0 < t < 12$, at which $Y(t) = 18$? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the mass of the yeast in the culture.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} Y(t)$
- ☐ interpretation

Solution:

$$\lim_{t \rightarrow \infty} Y(t) = \lim_{t \rightarrow \infty} \frac{600}{20+10e^{-0.05(t-12)}} = \frac{600}{20+0} = 30$$

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As time increases, the mass of the yeast in the culture approaches 30 milligrams.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 12^-} Y(t)$
- ☐ $\lim_{t \rightarrow 12^+} Y(t)$
- ☐ answer with justification

Solution

Because f is a continuous function and $f(12) = 20$, $\lim_{t \rightarrow 12^-} Y(t) = \lim_{t \rightarrow 12^-} f(t) = f(12) = 20$.

$$\lim_{t \rightarrow 12^+} Y(t) = \lim_{t \rightarrow 12^+} \frac{600}{20 + 10e^{-0.05(t-12)}} = \frac{600}{20 + 10} = 20$$

$$Y(12) = f(12) = 20$$

Yes, Y is continuous at $t = 12$ because $\lim_{t \rightarrow 12} Y(t) = Y(12)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

- ☐ answer with justification
- ☐ specific reference to *IVT*

Unit 1 AP Topics 15**Solution:**

Y is continuous for $0 \leq t \leq 12$.

$$Y(0) = 10 < 18 < 20 = Y(12)$$

By the Intermediate Value Theorem, there must be at least one time t , $0 \leq t \leq 12$, such that $Y(t) = 18$.

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47. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$N(t) = \begin{cases} f(t) & \text{for } 0 \leq t < 6 \\ 25t + 150 & \text{for } 6 \leq t < 8 \\ \frac{200+80t}{2+0.05t} & \text{for } t \geq 8 \end{cases}$$

The number of fish in a pond at time t years is modeled by the function N defined above, where f is a continuous function such that $f(0) = 80$.

- (a) Find $\lim_{t \rightarrow \infty} N(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} N(t)$ in the context of the problem.
- (b) Is the function N continuous at $t = 8$? Justify your answer.
- (c) The function N is continuous at $t = 6$. Is there a time t , for $0 \leq t \leq 6$, at which $N(t) = 250$? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the number of fish.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} N(t)$
- ☐ interpretation

Solution:

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$$\lim_{t \rightarrow \infty} N(t) = \lim_{t \rightarrow \infty} \frac{200+80t}{2+0.05t} = \frac{80}{0.05} = 1600$$

As time increases, the number of fish in the pond approaches 1600.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 8^-} N(t)$
- ☐ $\lim_{t \rightarrow 8^+} N(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 8^-} N(t) = \lim_{t \rightarrow 8^-} (25t + 150) = 350$$

$$\lim_{t \rightarrow 8^+} N(t) = \lim_{t \rightarrow 8^+} \frac{200+80t}{2+0.05t} = \frac{840}{2.4} = 350$$

$$N(8) = 350$$

Yes, N is continuous at $t = 8$ because

$$\lim_{t \rightarrow 8} N(t) = N(8).$$

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

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- ☐ answer with justification
- ☐ specific reference to IVT

Solution:

N is continuous for $0 \leq t \leq 6$.

$$N(0) = 80 < 250 < 300 = N(6)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 6$, such that

$$N(t) = 250.$$

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48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$R(t) = \begin{cases} 41\sqrt{\frac{t}{20}} & \text{for } 0 \leq t < 20 \\ \frac{100t^2 + 1000}{t^2 + 600} & \text{for } t \geq 20 \end{cases}$$

The total accumulated revenue a company has received up to time t is modeled by the function R defined above, where $R(t)$ is measured in millions of United States dollars and t is the time in years since 1980.

- Find $\lim_{t \rightarrow \infty} R(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} R(t)$ in the context of the problem.
- Is the function R continuous at $t = 20$? Justify your answer.
- The company's total accumulated expenses up to time t is modeled by the function E defined by $E(t) = 3\log_2(12t + 4)$, where $E(t)$ is measured in millions of United States dollars and t is the time in years since 1980. According to these models, is there a time t , for $0 \leq t \leq 5$, at which the total accumulated profit, $R(t) - E(t)$, is equal to 0? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the revenue of the company.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} R(t)$
- ☐ interpretation

Solution:

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$$\lim_{t \rightarrow \infty} R(t) = \lim_{t \rightarrow \infty} \frac{100t^2 + 1000}{t^2 + 600} = \frac{100}{1} = 100$$

As time increases, the total accumulated revenue of the company approaches \$100 million.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 20^-} R(t)$
- ☐ $\lim_{t \rightarrow 20^+} R(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 20^-} R(t) = \lim_{t \rightarrow 20^-} 41\sqrt{\frac{t}{20}} = 41\sqrt{\frac{20}{20}} = 41$$

$$\lim_{t \rightarrow 20^+} R(t) = \lim_{t \rightarrow 20^+} \frac{100t^2 + 1000}{t^2 + 600} = \lim_{t \rightarrow 20^+} \frac{100(t^2 + 10)}{t^2 + 600} = \frac{100(20^2 + 10)}{20^2 + 600} = \frac{100(410)}{1000} = 41$$

$$R(20) = 41$$

Yes, R is continuous at $t = 20$ because $\lim_{t \rightarrow 20} R(t) = R(20)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

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- ☐ answer with justification
- ☐ specific reference to IVT.

Solution:

R and E are continuous for $0 \leq t \leq 5$.

Therefore, the function defined by $R(t) - E(t)$ is continuous for $0 \leq t \leq 5$.

$$R(0) - E(0) = 41\sqrt{\frac{0}{20}} - 3\log_2(12 \cdot 0 + 4) = 0 - 6 = -6 < 0$$

$$\begin{aligned} R(5) - E(5) &= 41\sqrt{\frac{5}{20}} - 3\log_2(12 \cdot 5 + 4) \\ &= \frac{41}{2} - 3\log_2 64 = 20.5 - 18 > 0 \end{aligned}$$

$$R(0) - E(0) < 0 < R(5) - E(5)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 5$, such that $R(t) - E(t) = 0$.

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49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the x -axis such that, for time $t > 0$, its position is given by $x_P(t) = 6 - 4e^{-t}$.

Particle Q moves along the y -axis such that, for time $t > 0$, its velocity is given by $v_Q(t) = \frac{1}{t^2}$. At time $t = 1$, the position of particle Q is $y_Q(1) = 2$.

- (a) Find $v_P(t)$, the velocity of particle P at time t .
- (b) Find $a_Q(t)$, the acceleration of particle Q at time t . Find all times t , for $t > 0$, when the speed of particle Q is decreasing. Justify your answer.
- (c) Find $y_Q(t)$, the position of particle Q at time t .
- (d) As $t \rightarrow \infty$, which particle will eventually be farther from the origin? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that equates $x_P(t)$ with $v_P(t)$ does not earn the point.

An unlabeled response earns the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$v_P(t) = x_P'(t) = 4e^{-t}$$

Part B

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earning the first point is not necessary for a response to be eligible to earn the second or third points; however, the response must present an expression for $a_Q(t)$ to be eligible for third point.

A response earns the second point with either of the following statements: “ $v_Q(t)$ and $a_Q(t)$ have opposite signs” or “ $v_Q(t)$ and $a_Q(t)$ have the same sign.” This statement, however, must be consistent with $v_Q(t)$ and the presented expression for $a_Q(t)$.

A response must earn the second point to be eligible for the third point. The answer must be consistent with the presented justification. Furthermore, responses for which $a_Q(t) > 0$ for $t > 0$ must conclude that there is no time at which the speed of the particle is decreasing.

A response that indicates $v_Q(t) < 0$ does not earn the third point, even if the answer and justification are consistent with a reported sign of $a_Q(t)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $a_Q(t)$
- ☐ Considers signs of $a_Q(t)$ and $v_Q(t)$
- ☐ Answer with justification

Solution:

$$a_Q(t) = v_Q'(t) = \frac{-2}{t^3}$$

For $t > 0$, $a_Q(t) < 0$ and $v_Q(t) > 0$.

Because the velocity and acceleration have opposite signs, the speed of particle Q is decreasing for all $t > 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents $\int_1^t \frac{1}{t^2} dt$ (using the same variable as a limit and integrand function) does not earn the first point unless it is followed by an attempt at integration.

A response that presents either $\int \frac{1}{t^2} dt$ or $-\frac{1}{t}$ (with no integral) earns the first point. If the response continues and presents $2 = -1 + C$, then the response earns the second point.

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A response that presents only $y_Q(t) = -\frac{1}{t} + 3$ will earn all 3 points. Note that the right side of this equation suffices to earn all points. A response of $y_Q(t) = -\frac{1}{t} + C$, where $C \neq 3$, (with no additional supporting work) earns only the first point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integral
- ☐ Uses initial condition
- ☐ Answer

Solution:

$$y_Q(t) = y_Q(1) + \int_1^t \frac{1}{s^2} ds$$

$$= 2 - \left(\frac{1}{s} \Big|_1^t \right) = 2 - \frac{1}{t} + 1 = 3 - \frac{1}{t}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with an incorrect $y_Q(t)$ from part (c) is eligible for both points in part (d) provided $y_Q(t)$ is a non-constant function. The second point is earned for a consistent answer with reason, and limits correct for particle P and the presented $y_Q(t)$.

Responses that present statements such as “ $6 - 4e^{-t}$ approaches 6” or “ Q goes to 3” earn the first point and are eligible for the second point.

A response that treats ∞ as an input for $x_P(t)$ or $y_Q(t)$, such as “ $6 - 4e^{-\infty}$ ” or “ $3 - \frac{1}{\infty}$ ” is not eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ One correct limit
- ☐ Answer with reason

Unit 1 AP Topics 15**Solution:**

For particle P , $\lim_{t \rightarrow \infty} (6 - 4e^{-t}) = 6$.

For particle Q , $\lim_{t \rightarrow \infty} (3 - \frac{1}{t}) = 3$.

Because $6 > 3$, particle P will eventually be farther from the origin.

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50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the x -axis so that its position at time $t > 0$ is given by $x_P(t) = \frac{e^{2-t}-2t}{e^{2-t}+3t}$. A second particle, particle Q , also moves along the x -axis so that its position at time t is given by $x_Q(t) = t^3 - 3t^2 + 5$.

- Show that the velocity of particle P at time t is given by $v_P(t) = \frac{-5te^{2-t}-5e^{2-t}}{(e^{2-t}+3t)^2}$.
- At time $t = 2$, particle Q is at rest. At time $t = 2$, is particle P moving toward particle Q or away from particle Q ? Explain your reasoning.
- The acceleration of particle Q is given by $a_Q(t)$. Find the value of $a_Q(2)$.
- Describe the position of particle P and the position of particle Q as t approaches infinity. Show the work that leads to your answers.

Part A

The first point is earned for successful demonstration of the quotient rule.

The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ quotient rule
- ☐ verification

Solution:

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$$v_P(t) = x'(t) = \frac{(-e^{2-t}-2)(e^{2-t}+3t) - (e^{2-t}-2t)(-e^{2-t}+3)}{(e^{2-t}+3t)^2}$$

$$= \frac{(-e^{4-2t}-3te^{2-t}-2e^{2-t}-6t) - (-e^{4-2t}+3e^{2-t}+2te^{2-t}-6t)}{(e^{2-t}+3t)^2} = \frac{-5te^{2-t}-5e^{2-t}}{(e^{2-t}+3t)^2}$$

Part B

The response is eligible for the third point if either of the first 2 points is earned and the answer with reason is consistent with the previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $x_P(2)$ and $x_Q(2)$
- ☐ $v_P(2) < 0$
- ☐ answer with explanation

Solution:

$$x_P(2) = \frac{1-4}{1+6} = -\frac{3}{7};$$

$$x_Q(2) = 1$$

$$v_P(2) = \frac{-10-5}{7^2} = -\frac{15}{49} < 0$$

Because $x_P(2) < x_Q(2)$, particle P is to the left of particle Q at time $t = 2$.

Because $v_P(2) < 0$, P is moving to the left at time $t = 2$.

Because particle Q is at rest at time $t = 2$, particle P is moving away from particle Q at time $t = 2$.

Part C

Supporting work is not required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
---	---

The student response accurately includes a correct value of $a_Q(2)$

$$v_Q(t) = x_Q'(t) = 3t^2 - 6t$$

$$a_Q(t) = v_Q'(t) = 6t - 6$$

$$a_Q(2) = 12 - 6 = 6$$

Part D

The first point requires showing that an indeterminate form $\frac{\infty}{\infty}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $\frac{\infty}{\infty}$ " or not using limit notation) impacts the first point.

For the second and third points, at most 1 out of 2 points may be earned for both correct limits without a connection to the interpretation of position.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ description of position of particle P
- ☐ description of position of particle Q

Solution:

$$\lim_{t \rightarrow \infty} e^{2-t} - 2t = -\infty$$

$$\lim_{t \rightarrow \infty} e^{2-t} + 3t = \infty$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow \infty} x_P(t) = \lim_{t \rightarrow \infty} \frac{e^{2-t} - 2t}{e^{2-t} + 3t} = \lim_{t \rightarrow \infty} \frac{-e^{2-t} - 2}{-e^{2-t} + 3} = -\frac{2}{3}$$

Therefore, particle P approaches $x = -\frac{2}{3}$ as t approaches infinity.

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$$\lim_{t \rightarrow \infty} x_Q(t) = \lim_{t \rightarrow \infty} (t^3 - 3t^2 + 5) = \infty$$

Therefore, the position of particle Q grows without bound as t approaches infinity.

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51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle moves along the x -axis so that its position at time $t > 0$ is given by $x(t) = \frac{t^2 - 9}{3t^2 + 8}$.

- Show that the velocity of the particle at time t is given by $v(t) = \frac{70t}{(3t^2 + 8)^2}$.
- Is the particle moving toward the origin or away from the origin at time $t = 2$? Give a reason for your answer.
- The acceleration of the particle is given by $a(t)$. Write an expression for $a(t)$, and find the value of $a(2)$.
- What position does the particle approach as t approaches infinity?

Part A

The first point is earned for successful demonstration of the quotient rule.

The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ quotient rule
- ☐ verification

Solution:

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$$v(t) = x'(t) = \frac{2t(3t^2+8) - (t^2-9)6t}{(3t^2+8)^2}$$

$$= \frac{6t^3+16t-6t^3+54t}{(3t^2+8)^2} = \frac{70t}{(3t^2+8)^2}$$

Part B

The response is eligible for the third point if either of the first 2 points is earned and the answer with reason is consistent with the previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $x(2) < 0$
- ☐ $v(2) > 0$
- ☐ answer with reason

Solution:

$$x(2) = \frac{-5}{20} = -\frac{1}{4} < 0$$

$$v(2) = \frac{140}{20^2} = \frac{7}{20} > 0$$

Because $x(2) < 0$, the particle is to the left of the origin at time $t = 2$.

Because $v(2) > 0$, the particle is moving to the right at time $t = 2$.

Therefore, the particle is moving toward the origin at time $t = 2$.

Part C

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error.

A simplified answer is not required for the third point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

Solution:

$$a(t) = v'(t) = \frac{70(3t^2+8)^2 - 70t \cdot 2(3t^2+8) \cdot 6t}{(3t^2+8)^4}$$

$$= \frac{70(3t^2+8) - 70 \cdot 12t^2}{(3t^2+8)^3} = \frac{70(8-9t^2)}{(3t^2+8)^3}$$

$$a(2) = \frac{70(8-9(2)^2)}{(3(2)^2+8)^3} = \frac{70(-28)}{20^3} = -\frac{49}{200}$$

Part D

Supporting work is not required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct position.

Solution:

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} \frac{t^2-9}{3t^2+8} = \frac{1}{3}$$

The particle approaches $x = \frac{1}{3}$ as t approaches infinity.

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52. If f is a function such that $\lim_{x \rightarrow \infty} f(x) = 0$, which of the following could be an expression for $f(x)$?

(A) $\frac{x^2}{x^2 + 4}$

(B) $\frac{\sin x}{x}$

(C) $\cos x$

(D) e^x

**Answer B**

Correct. Because $-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$ for $x > 0$ and $\lim_{x \rightarrow \infty} \left(-\frac{1}{x}\right) = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$, it follows that $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$ by the squeeze theorem.

53. $\lim_{x \rightarrow \infty} \frac{x+7}{\sqrt{2x^2+49}}$ is

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) 1

**Answer C**

Correct. The limit can be evaluated using algebraic manipulation, as follows.

$$\lim_{x \rightarrow \infty} \frac{x+7}{\sqrt{2x^2+49}} = \lim_{x \rightarrow \infty} \frac{x(1+\frac{7}{x})}{\sqrt{x^2(2+\frac{49}{x^2})}} = \lim_{x \rightarrow \infty} \frac{x(1+\frac{7}{x})}{x\sqrt{2+\frac{49}{x^2}}} = \lim_{x \rightarrow \infty} \frac{1+\frac{7}{x}}{\sqrt{2+\frac{49}{x^2}}} = \frac{1+0}{\sqrt{2+0}} = \frac{1}{\sqrt{2}}$$

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54. $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2+27}}{\sqrt{x^2+27x+3}}$ is

(A) 2

(B) 3

(C) 4

(D) infinite

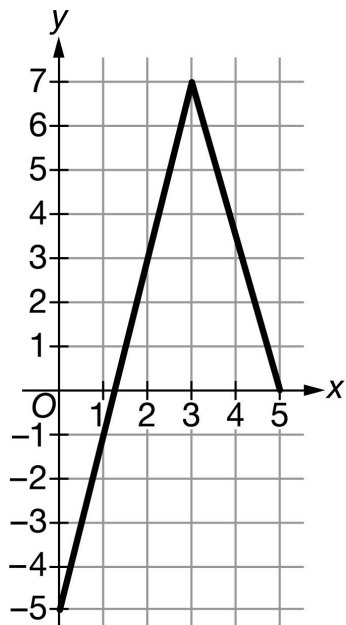
**Answer A**

Correct. The limit can be found by rearranging the expression into an equivalent form, as follows.

$$\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2+27}}{\sqrt{x^2+27x+3}} = \lim_{x \rightarrow \infty} \frac{\sqrt{4x^2\left(1+\frac{27}{4x^2}\right)}}{\sqrt{x^2\left(1+\frac{27}{x}+\frac{3}{x^2}\right)}} = \lim_{x \rightarrow \infty} \frac{2x\sqrt{1+\frac{27}{4x^2}}}{x\sqrt{1+\frac{27}{x}+\frac{3}{x^2}}} = 2 \lim_{x \rightarrow \infty} \frac{\sqrt{1+\frac{27}{4x^2}}}{\sqrt{1+\frac{27}{x}+\frac{3}{x^2}}} = 2 \frac{\sqrt{1+0}}{\sqrt{1+0+0}} = 2$$

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55.

Graph of f

The graph of a function f is shown above.

If g is the function defined by $g(x) = \frac{x^2+1}{f(x)}$, what is the value of $g'(2)$?

(A) $-\frac{8}{9}$

(B) $\frac{1}{9}$

(C) 1

(D) $\frac{32}{9}$

**Answer A**

Correct. The derivative of g is found using the quotient rule.

$$g'(x) = \frac{2xf(x) - f'(x)(x^2+1)}{(f(x))^2}$$

The graph of f is used to determine that $f(2) = 3$ and $f'(2) = \frac{7-3}{3-2} = 4$. Then

$$g'(2) = \frac{4f(2) - f'(2)(5)}{(f(2))^2} = \frac{(4)(3) - (4)(5)}{9} = -\frac{8}{9}.$$

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56. The function g is defined for all positive x , and $0 < g(x) < x$. Of the following, $\lim_{x \rightarrow \infty} \frac{x}{g(x)}$ could not be
- (A) 0
- (B) 1
- (C) 3
- (D) ∞

Answer C

Incorrect. $\lim_{x \rightarrow \infty} \frac{x}{g(x)}$ could equal 3. For example, if $g(x) = \frac{x}{3}$, then $0 < g(x) < x$ for all $x > 0$ and $\lim_{x \rightarrow \infty} \frac{x}{g(x)} = \lim_{x \rightarrow \infty} \frac{x}{\frac{x}{3}} = \lim_{x \rightarrow \infty} 3 = 3$.

57. If $f(t) = \frac{91}{18+25e^{-0.4t}}$, then $\lim_{t \rightarrow \infty} f(t)$ is
- (A) 0
- (B) $\frac{91}{43}$
- (C) $\frac{91}{18}$
- (D) nonexistent

Answer C

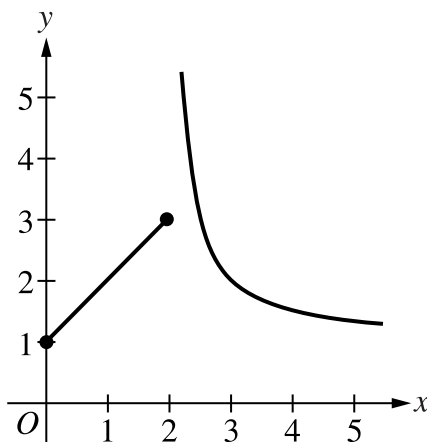
Correct. The limit of a quotient can be found using limit theorems.

$$\lim_{t \rightarrow \infty} (18 + 25e^{-0.4t}) = 18 + 25 \lim_{t \rightarrow \infty} e^{-0.4t} = 18 + 25 \cdot 0 = 18$$

Since the limit of the denominator is not 0, $\lim_{t \rightarrow \infty} \frac{91}{18+25e^{-0.4t}} = \frac{91}{\lim_{t \rightarrow \infty} (18+25e^{-0.4t})} = \frac{91}{18}$.

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58.

Graph of f

The figure shows the graph of a piecewise function f with domain $[0, \infty)$. The function f has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 1$. Which of the following statements are true?

- I. $\lim_{x \rightarrow 2^-} f(x)$ exists.
- II. $\lim_{x \rightarrow 2^+} f(x)$ exists.
- III. $\lim_{x \rightarrow \infty} f(x) = \infty$

(A) I only



(B) I and II only

(C) I and III only

(D) II and III only

Answer A

Correct. Only statement I is true.

Statement I is true because $\lim_{x \rightarrow 2^-} f(x) = 3$.

Statement II is false because $\lim_{x \rightarrow 2^-} f(x) = 3$ and $\lim_{x \rightarrow 2^+} f(x) = \infty$, so $\lim_{x \rightarrow 2} f(x)$ does not exist, since the limit from the right does not equal the limit from the left.

Statement III is false because $\lim_{x \rightarrow \infty} f(x) = 1$, since the function f has a horizontal asymptote at $y = 1$.

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59. Let f be the function given by $f(x) = \frac{e^{2x}}{1-e^{2x}}$. Which of the following describes all asymptotes of the graph of f ?
- (A) No vertical asymptotes and two horizontal asymptotes
- (B) One vertical asymptote and no horizontal asymptotes
- (C) One vertical asymptote and two horizontal asymptotes ✓
- (D) Two vertical asymptotes and two horizontal asymptotes

Answer C

Correct. The graph of f has only one vertical asymptote because $1 - e^{2x} = 0$ only at $x = 0$ where

$$\lim_{x \rightarrow 0^-} \frac{e^{2x}}{1-e^{2x}} = \infty \text{ and } \lim_{x \rightarrow 0^+} \frac{e^{2x}}{1-e^{2x}} = -\infty.$$

The graph has two horizontal asymptotes (at $y = 0$ and $y = -1$) because $\lim_{x \rightarrow -\infty} \frac{e^{2x}}{1-e^{2x}} = \frac{0}{1} = 0$ and

$$\lim_{x \rightarrow +\infty} \frac{e^{2x}}{1-e^{2x}} = \lim_{x \rightarrow +\infty} \frac{e^{2x}}{e^{2x}(e^{-2x}-1)} = \lim_{x \rightarrow +\infty} \frac{1}{e^{-2x}-1} = \frac{1}{-1} = -1.$$

60. Let f be the function given by $f(x) = \frac{(\cos x)e^{2x}-1}{e^{2x-1}+2}$. What are all horizontal asymptotes to the graph of f ?
- (A) $y = -\frac{1}{2}$ only ✓
- (B) $y = e$ only
- (C) $y = -\frac{1}{2}$ and $y = e$
- (D) The graph of f has no horizontal asymptotes.

Answer A

Correct. Since $\lim_{x \rightarrow -\infty} e^{2x} = 0$ and $\lim_{x \rightarrow -\infty} e^{2x-1} = 0$, $\lim_{x \rightarrow -\infty} f(x) = \frac{0-1}{0+2} = -\frac{1}{2}$. For large positive values of x ,

$$f(x) \approx \frac{(\cos x)e^{2x}}{e^{2x-1}} = \frac{(\cos x)e^{2x}}{e^{2x}(e^{-1})} = \frac{\cos x}{\frac{1}{e}} = e \cos x.$$

Therefore, $\lim_{x \rightarrow \infty} f(x)$ does not exist. As $x \rightarrow -\infty$, the graph of f has a horizontal asymptote at $y = -\frac{1}{2}$. As $x \rightarrow \infty$, the graph of f has no horizontal asymptote.

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61. Let f be the function defined by $f(x) = \frac{2x+3}{x+1}$. Which of the following statements are true?

- I. The graph of f has a horizontal asymptote at $y = 2$ because $\lim_{x \rightarrow \infty} f(x) = 2$.
 - II. The graph of f has a horizontal asymptote at $y = 2$ because $\lim_{x \rightarrow -\infty} f(x) = 2$.
 - III. The graph of f has a vertical asymptote at $x = -1$ because $\lim_{x \rightarrow -1^+} f(x) = \infty$.
- (A) I only
(B) III only
(C) I and II only
(D) I, II, and III

**Answer D**

Correct. $\lim_{x \rightarrow \pm\infty} \frac{2x+3}{x+1} = \lim_{x \rightarrow \pm\infty} \frac{2+\frac{3}{x}}{1+\frac{1}{x}} = \frac{2+0}{1+0} = 2$ and therefore the graph has a horizontal asymptote at $y = 2$ both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. As x approaches -1 from the right, the denominator approaches 0 and the numerator approaches 1. Therefore, the fraction will become unbounded as $x \rightarrow -1^+$, and so the graph of f will have a vertical asymptote at $x = -1$.

62. The population on an island is modeled by $P(t) = \frac{6000}{40+60e^{-0.03t}}$ for $t \geq 0$, where $P(t)$ is the number of people on the island after t years. What is $\lim_{t \rightarrow \infty} P(t)$?

- (A) 60
(B) 100
(C) 150
(D) 6000

**Answer C**

Correct. $\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} \frac{6000}{40+60e^{-0.03t}} = \frac{6000}{40+60\lim_{t \rightarrow \infty} e^{-0.03t}} = \frac{6000}{40+60 \cdot 0} = 150$

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63. Let f be the function defined by $f(x) = \frac{3x^{20}}{4e^x + 8x^{20}}$ for $x > 0$. Which of the following is a horizontal asymptote to the graph of f ?
- (A) $y = 0$ ✓
- (B) $y = \frac{3}{8}$
- (C) $y = \frac{3}{4}$
- (D) There is no horizontal asymptote to the graph of f .

Answer A

Correct. To determine the horizontal asymptote to the graph of f , $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{3x^{20}}{4e^x + 8x^{20}}$ is evaluated. The relative magnitude of e^x dominates that of x^{20} in the denominator as $x \rightarrow \infty$. Therefore, for large values of x , the fraction behaves like $\frac{3x^{20}}{4e^x}$. Since the exponential function grows much faster than the polynomial as $x \rightarrow \infty$, the limit will be 0.

64. Let f be the function defined by $f(x) = \frac{2-x+3x^2+5x^3-7x^4}{x^4-2x^3-5x^2+2x-3}$ for $x > 0$. Which of the following is a horizontal asymptote to the graph of f ?
- (A) $y = -7$ ✓
- (B) $y = 2$
- (C) $y = 7$
- (D) There is no horizontal asymptote to the graph of f .

Answer A

Correct. To determine the horizontal asymptote to the graph of f ,

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2-x+3x^2+5x^3-7x^4}{x^4-2x^3-5x^2+2x-3}$$

is evaluated. The term in the numerator with the highest degree is $-7x^4$, and the term in the denominator with the highest degree is x^4 . The limit as x approaches infinity of this rational function can be found by considering the ratio of these two terms.

$$\lim_{x \rightarrow \infty} \frac{2-x+3x^2+5x^3-7x^4}{x^4-2x^3-5x^2+2x-3} = \lim_{x \rightarrow \infty} \frac{-7x^4}{x^4} = \lim_{x \rightarrow \infty} -7 = -7$$

Therefore, $y = -7$ is a horizontal asymptote to the graph of f .

Unit 1 AP Topics 15

65. Let f be the function given by $f(x) = \frac{1+e^x \sin x}{e^{x-1}-1}$. What are all horizontal asymptotes of the graph of f ?

- (A) $y = -1$ only ✓
- (B) $y = 1$ only
- (C) $y = -1$ and $y = 1$
- (D) The graph of f has no horizontal asymptotes.

Answer A

Correct. Since $\lim_{x \rightarrow -\infty} \frac{1+e^x \sin x}{e^{x-1}-1} = \frac{1}{-1} = -1$, the line $y = -1$ is a horizontal asymptote. For large positive values of x , $f(x) \approx \frac{e^x \sin x}{e^{x-1}} = e \sin x$. Therefore, $\lim_{x \rightarrow \infty} f(x)$ does not exist and the graph does not approach a horizontal asymptote as $x \rightarrow \infty$.

66. Let f be the function defined by $f(x) = \frac{1-5x-2x^2}{3x^2+7}$ for $x > 0$. Which of the following is a horizontal asymptote to the graph of f ?

- (A) $y = -\frac{2}{3}$ ✓
- (B) $y = \frac{1}{3}$
- (C) $y = \frac{2}{3}$
- (D) There is no horizontal asymptote to the graph of f .

Answer A

Correct. To determine the horizontal asymptote to the graph of f , $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{1-5x-2x^2}{3x^2+7}$ is evaluated. The term in the numerator with the highest power is $-2x^2$, and the term in the denominator with the highest power is $3x^2$. The limit as x approaches infinity of this rational function can be found by considering the ratio of these two terms. $\lim_{x \rightarrow \infty} \frac{1-5x-2x^2}{3x^2+7} = \lim_{x \rightarrow \infty} \frac{-2x^2}{3x^2} = \lim_{x \rightarrow \infty} \frac{-2}{3} = -\frac{2}{3}$

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67. Let f be the function defined by $f(x) = \frac{2^x+5}{e^x+1}$ for $x > 0$. Which of the following is a horizontal asymptote to the graph of f ?

- (A) $y = 0$ ✓
(B) $y = \frac{2}{e}$
(C) $y = 1$
(D) There is no horizontal asymptote to the graph of f .

Answer A

Correct. To determine the horizontal asymptote to the graph of f , $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2^x+5}{e^x+1}$ is evaluated. For large values of x , $\frac{2^x+5}{e^x+1}$ behaves like $\frac{2^x}{e^x}$.

$$\lim_{x \rightarrow \infty} \frac{2^x+5}{e^x+1} = \lim_{x \rightarrow \infty} \frac{2^x}{e^x} = \lim_{x \rightarrow \infty} \left(\frac{2}{e}\right)^x.$$

Since $0 < \frac{2}{e} < 1$, $\lim_{x \rightarrow \infty} \left(\frac{2}{e}\right)^x = 0$.

68. Let f be the function defined by $f(x) = \frac{3x+5}{x+2}$. Which of the following statements are true?

- I. The graph of f has a horizontal asymptote at $y = 3$ because $\lim_{x \rightarrow \infty} f(x) = 3$.
II. The graph of f has a horizontal asymptote at $y = 3$ because $\lim_{x \rightarrow -\infty} f(x) = 3$.
III. The graph of f has a vertical asymptote at $x = -2$ because $\lim_{x \rightarrow -2^+} f(x) = -\infty$.
- (A) I only
(B) III only
(C) I and II only
(D) I, II, and III ✓

Answer D

Correct. $\lim_{x \rightarrow \pm\infty} \frac{3x+5}{x+2} = \lim_{x \rightarrow \pm\infty} \frac{3+\frac{5}{x}}{1+\frac{2}{x}} = \frac{3+0}{1+0} = 3$ and therefore the graph has a horizontal asymptote at $y = 3$ both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. As x approaches -2 from the right, the denominator approaches 0 and the numerator approaches -1 . Therefore, the fraction will become unbounded as $x \rightarrow -2^+$, and so the graph of f will have a vertical asymptote at $x = -2$.

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69. The population on an island is modeled by $P(t) = \frac{5000}{20+30e^{-0.03t}}$ for $t \geq 0$, where $P(t)$ is the number of people on the island after t years. What is $\lim_{t \rightarrow \infty} P(t)$?
- (A) 100
(B) $\frac{500}{3}$
(C) 250
(D) 5000

Answer C

Correct. $\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} \frac{5000}{20+30e^{-0.03t}} = \frac{5000}{20+30 \lim_{t \rightarrow \infty} e^{-0.03t}} = \frac{5000}{20+30 \cdot 0} = \frac{5000}{20} = 250$

70. Let f be the function defined by $f(x) = \frac{5x^{20}}{8e^x + 9x^{20}}$ for $x > 0$. Which of the following is a horizontal asymptote to the graph of f ?
- (A) $y = 0$
(B) $y = \frac{5}{9}$
(C) $y = \frac{5}{8}$
(D) There is no horizontal asymptote to the graph of f .

Answer A

Correct. To determine the horizontal asymptote to the graph of f , $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{5x^{20}}{8e^x + 9x^{20}}$ is evaluated. The relative magnitude of e^x dominates that of x^{20} in the denominator as $x \rightarrow \infty$. Therefore, for large values of x , the fraction behaves like $\frac{5x^{20}}{8e^x}$. Since the exponential function grows much faster than the polynomial as $x \rightarrow \infty$, the limit will be 0.

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71. Let f be the function defined by $f(x) = \frac{3^x+2}{e^{2x}+1}$ for $x > 0$. Which of the following is a horizontal asymptote to the graph of f ?

(A) $y = 0$



(B) $y = \frac{3}{e^2}$

(C) $y = 1$

(D) There is no horizontal asymptote to the graph of f .

Answer A

Correct. To determine the horizontal asymptote to the graph of f ,

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{3^x+2}{e^{2x}+1}$$

is evaluated. For large values of x , $\frac{3^x+2}{e^{2x}+1}$ behaves like $\frac{3^x}{e^{2x}}$.

$$\lim_{x \rightarrow \infty} \frac{3^x+2}{e^{2x}+1} = \lim_{x \rightarrow \infty} \frac{3^x}{e^{2x}} = \lim_{x \rightarrow \infty} \left(\frac{3}{e^2}\right)^x$$

Since $0 < \frac{3}{e^2} < 1$, $\lim_{x \rightarrow \infty} \left(\frac{3}{e^2}\right)^x = 0$. Therefore, $y = 0$ is a horizontal asymptote to the graph of f .

72. $\lim_{x \rightarrow \infty} \frac{10-6x^2}{5+3e^x}$ is

(A) -2

(B) 0



(C) 2

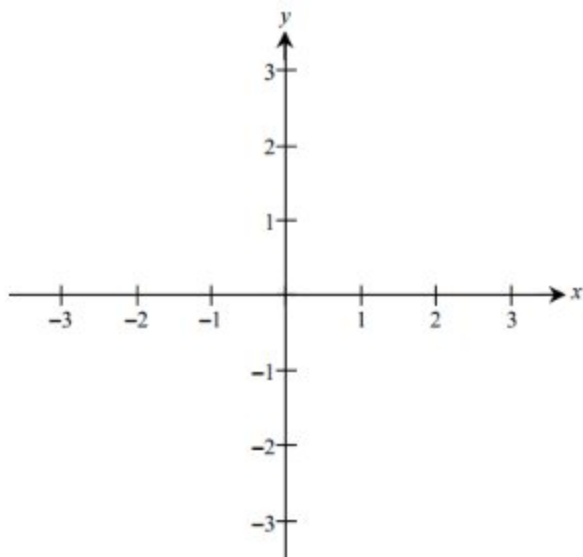
(D) nonexistent

Answer B

Correct. The numerator of $\frac{10-6x^2}{5+3e^x}$ is a translated power function and the denominator is a translated exponential function. Since the exponential function e^x grows faster than the power function x^2 , the relative magnitude of the denominator compared to the numerator will result in this expression converging to 0 as x goes to infinity.

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73. Let f be the function defined by $f(x) = xe^{1-x}$ for all real numbers x .
- Find each interval on which f is increasing.
 - Find the range of f .
 - Find the x -coordinate of each point of inflection of the graph of f .
 - Using the results found in parts (a), (b), and (c), sketch the graph of f in the xy -plane provided below. (Indicate all intercepts.)



Part A

1 point is earned for derivative

1 point is earned for answer

$$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$$

f increases on $(-\infty, 1]$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for derivative

1 point is earned for answer

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$$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$$

f increases on $(-\infty, 1]$

Part B

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range: $(-\infty, 1]$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range: $(-\infty, 1]$

Part C

1 point is earned for $f''(x)$ for student's $f(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

Point of inflection at $x = 2$

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✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f''(x)$ for student's $f'(x)$

1 point is earned for answer

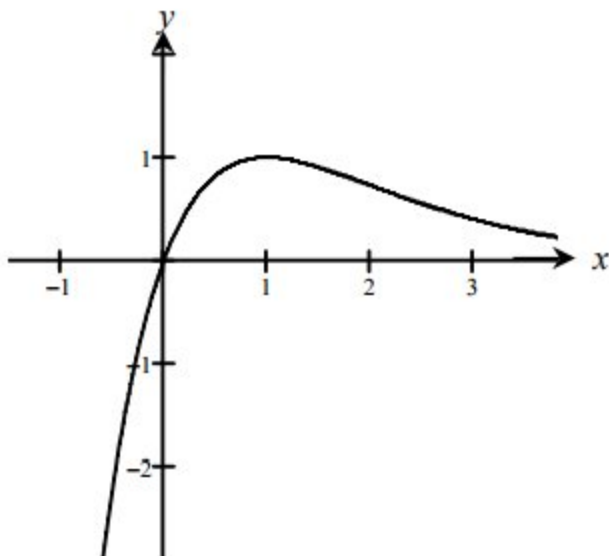
$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

Point of inflection at $x = 2$

Part D

3 points are earned for answer without mistake

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.



✓

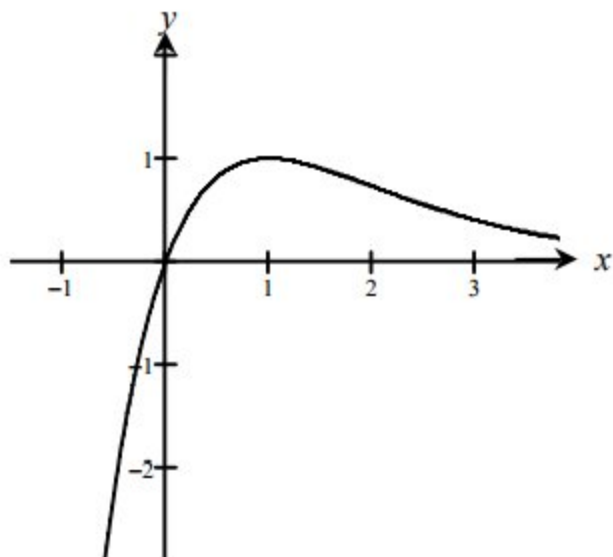
0	1	2	3
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The student response earns all of the following points:

3 points are earned for answer without mistake

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1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.



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74. Let f be the function given by $f(x) = \frac{|x|-2}{x-2}$.

- (a) Find all the zeros of f .
- (b) Find $f'(1)$.
- (c) Find $f'(-1)$.
- (d) Find the range of f .

Part A

1 point is earned for $x = -2$

$$f(x) = 0 \Leftrightarrow |x| = 2, x \neq 2$$

$$x = -2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for $x = -2$

$$f(x) = 0 \Leftrightarrow |x| = 2, x \neq 2$$

$$x = -2$$

Part B

1 point is earned for $f'(1) = 0$

For $x \geq 0, x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f'(1) = 0$



0	1
---	---

The student response earns all of the following points:

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1 point is earned for $f'(1) = 0$

For $x \geq 0, x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f'(1) = 0$

Part C

4 points are earned if solves for $f'(x)$ and $f'(-1)$

$$\text{For } x < 0, f(x) = \frac{-x-2}{x-2} = \frac{-(x+2)}{x-2}$$

$$f'(x) = \frac{(x-2)(-1) - (-x-2)(1)}{(x-2)^2} = \frac{4}{(x-2)^2}$$

$$\therefore f'(-1) = \frac{4}{9}$$



0	1	2	3	4
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The student response earns all of the following points:

4 points are earned if solves for $f'(x)$ and $f'(-1)$

$$\text{For } x < 0, f(x) = \frac{-x-2}{x-2} = \frac{-(x+2)}{x-2}$$

$$f'(x) = \frac{(x-2)(-1) - (-x-2)(1)}{(x-2)^2} = \frac{4}{(x-2)^2}$$

$$\therefore f'(-1) = \frac{4}{9}$$

Part D

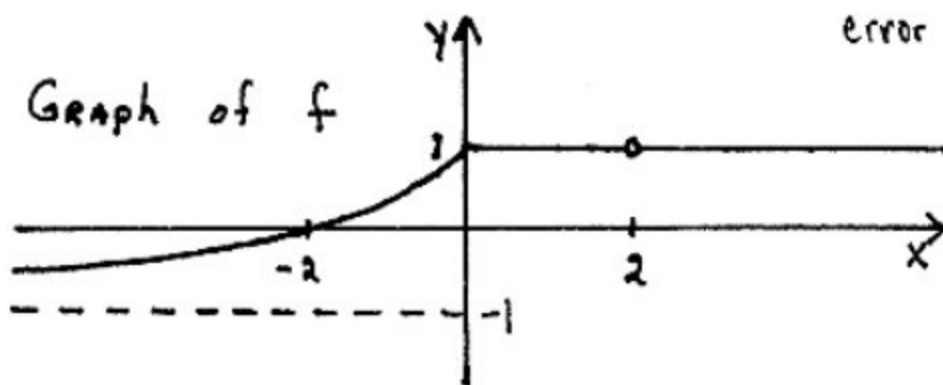
1 point is earned for least upper bound (lub) $y = 1$

1 point is earned for greatest lower bound (glb) $y = -1$

1 point is earned for correct inequalities at -1 and 1

$$-1 < y \leq 1$$

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✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for least upper bound (lub) $y = 1$

1 point is earned for greatest lower bound (glb) $y = -1$

1 point is earned for correct inequalities at -1 and 1

$-1 < y \leq 1$

